

Introduction to Biometric Technologies and Applications (IV)

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Course: Biometric Recognition
DI4025

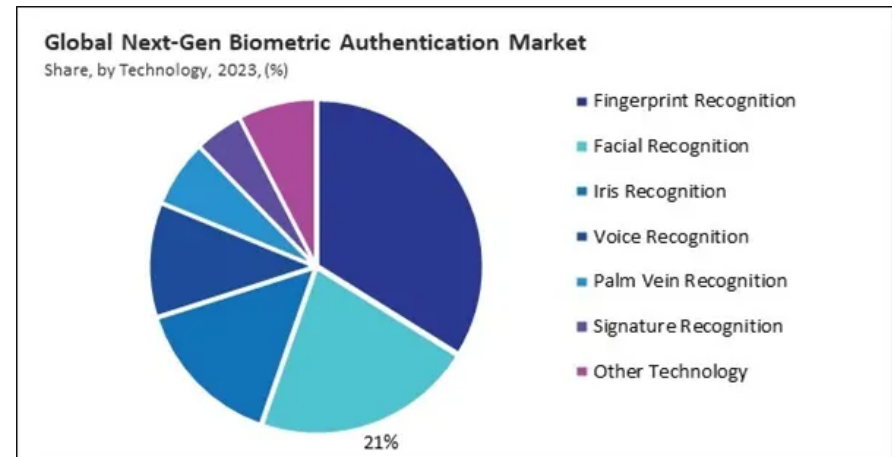
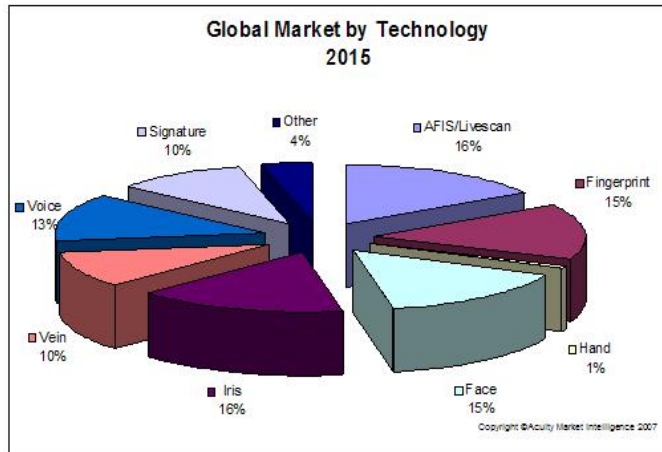
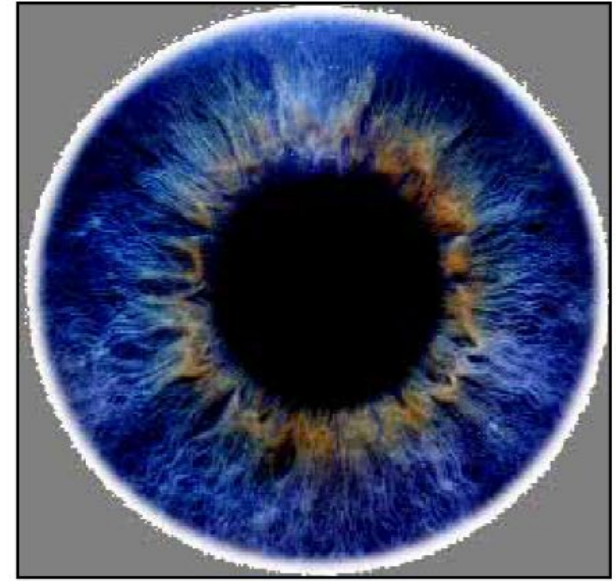
Outline

Overview of Main Biometric Modalities

- Fingerprint
- Face
- Iris

Other Biometric Modalities

The three Most Popular Biometric Traits



Outline

Overview of Main Biometric Modalities

- **Fingerprint**
- Face
- Iris

Other Biometric Modalities

Why fingerprints?



- Highly distinctive and unique
- Do not change during the lifetime of a person
- **Publicly accepted as reliable (evidence in a court of law)**
- **Identical twins have different fingerprints**



An impression on a
Palestinian lamp (400 B.C.)

Fingerprint Recognition Milestones

1880

Henry Faulds

Article on fingerprints published in Nature



1901

Scotland Yard

Adopted Galton/Henry system of classification



1963

Mitchell Trauring

First paper on automatic fingerprint matching

2004

DHS

US-VISIT

2009

UIDAI

India started issuing 12-digit UID no. to its residents



AD 600

China

Fingerprint to seal contracts and legal documents

1893

Argentina

First use of fingerprint as forensic evidence

1924

FBI

Set up fingerprint identification division

1970s

FBI

Initiation of AFIS

2008

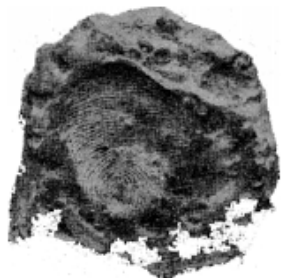
FBI

NGI

~1000-2000 B.C.

Ancient Babylon

Finger mark on clay seal



1892

Juan Vucetich

Rolled ink on paper



1990s

Live-scan technique

- Optical sensor
- Solid-state sensor
- Ultrasound sensor

2003

US-VISIT



1997

Thomson-CSF

Swipe sensor



~1995

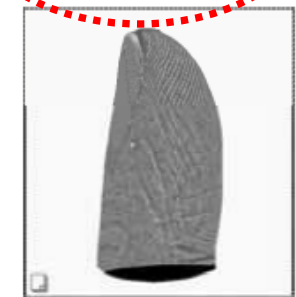


Siemens ID mouse

2005

TBS

Touchless 3D sensor



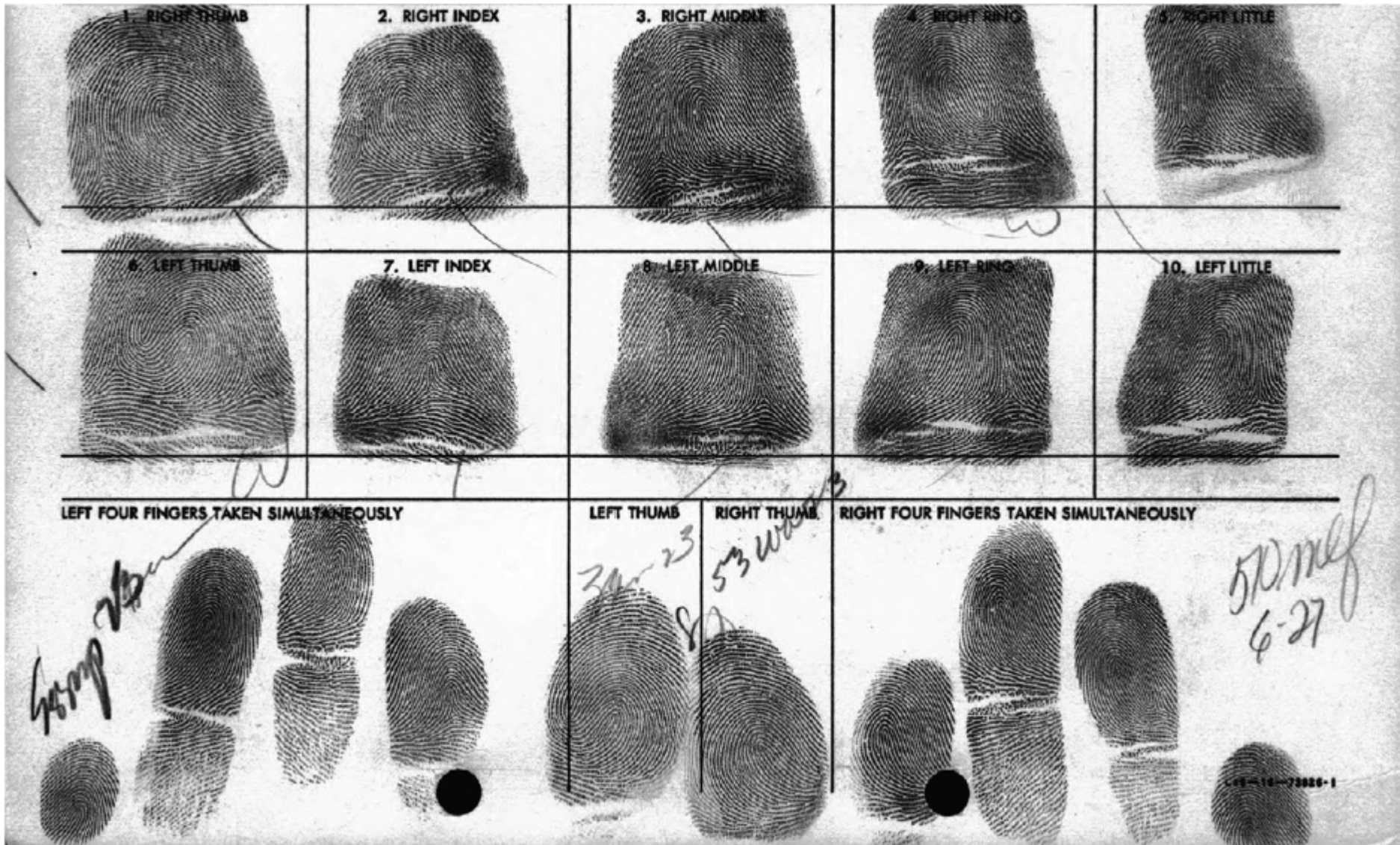
Fingerprint acquisition

- Off-line acquisition
 - Ink technique
 - Latent fingerprints
- On-line acquisition
 - Optical sensors
 - Silicon-based sensors
 - Ultrasound
 - Fingerphoto
 - Emerging (OCT, Organic)



Off-line – ink technique

- Rolled or plain



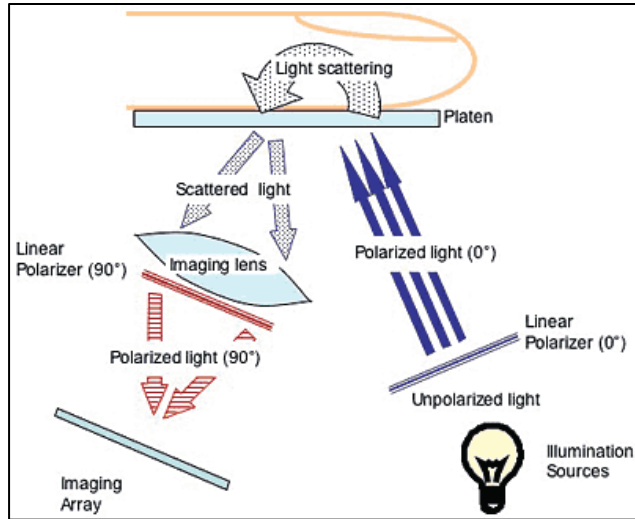
Off-line – latent fingerprints

- Lifted from surfaces of objects touched inadvertently
- Extracted using a variety of methods

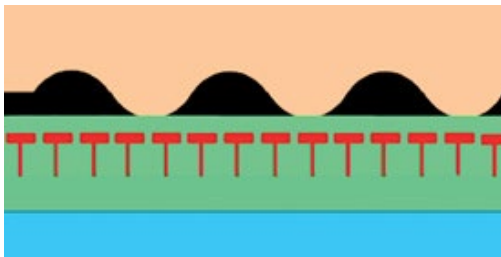


On-line scanners – single finger

- Optical sensors: measure light reflection



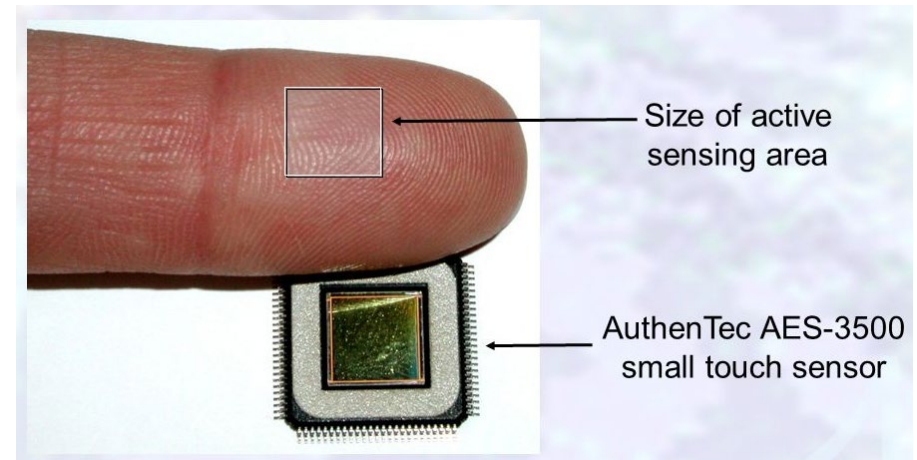
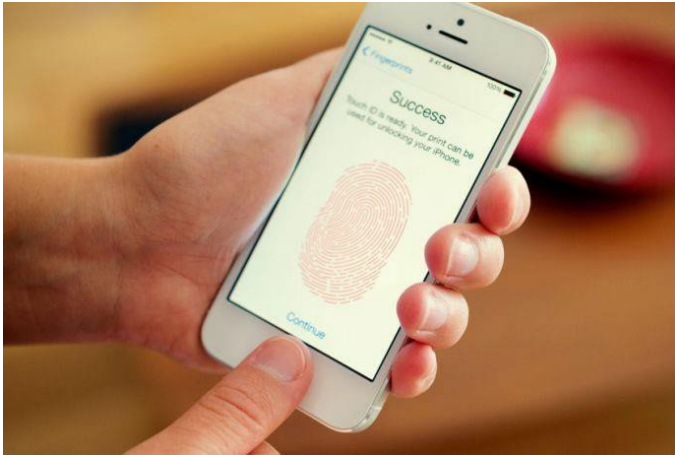
- Capacitive sensors: measure skin conductivity



On-line scanners – multi finger



Problems with small area sensors



Fingerprint vs Fingerphoto

Fingerphoto



Low contrast
Complex background
Natural lighting
Shadows

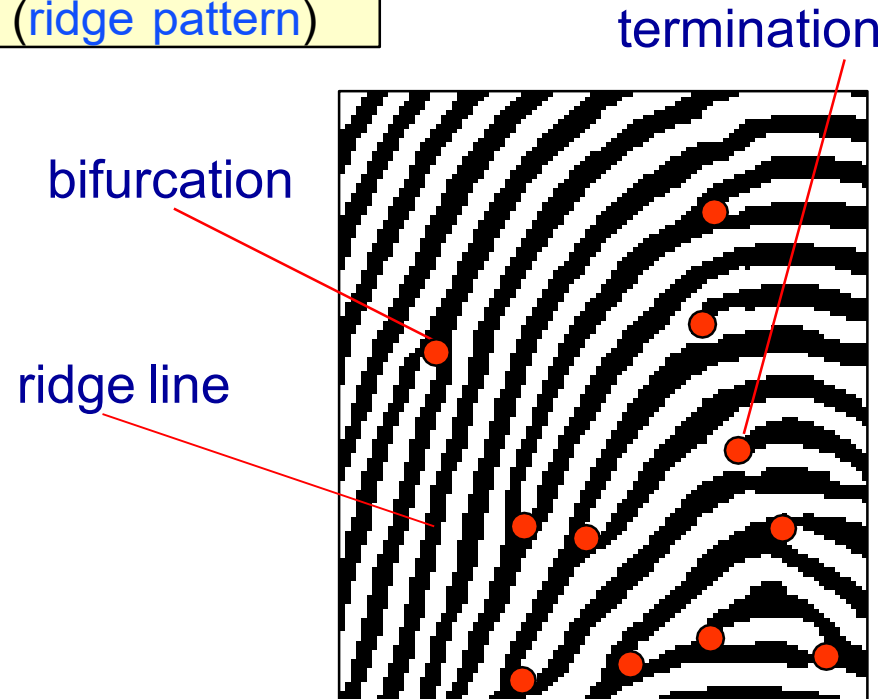
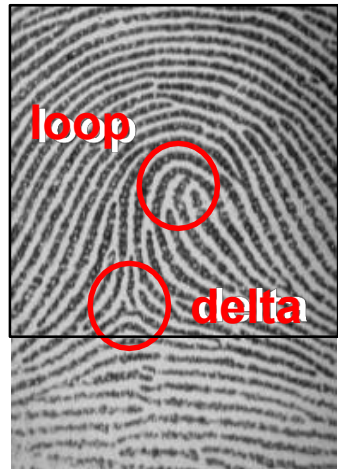


Fingerprint

Fingerprint Anatomy

A fingerprint is composed of a set of lines (**ridge lines**), which mainly flow parallel, making a pattern (**ridge pattern**)

Sometimes the ridge lines produce local **macro-singularities**, called **whorl** (O), **loop** (U) and **delta** (Δ)



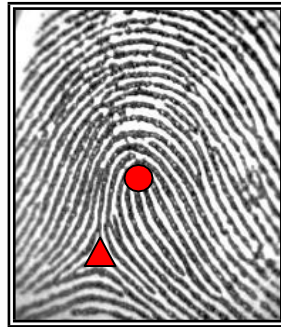
The **minutiae**, or **Galton's** characteristics, are determined by the **termination** or the **bifurcation** of the ridge lines

Fingerprint Anatomy

arch

tented arch

right loop



left loop

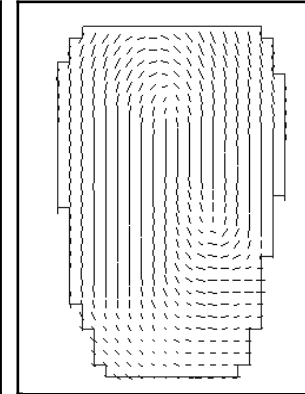


whorl

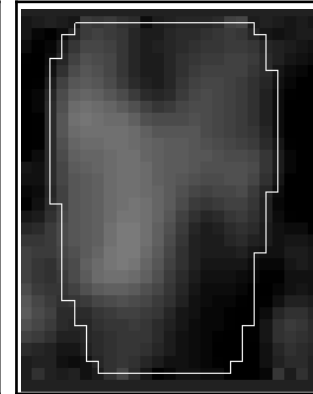
The five main fingerprint classes



fingerprint



orientation
image

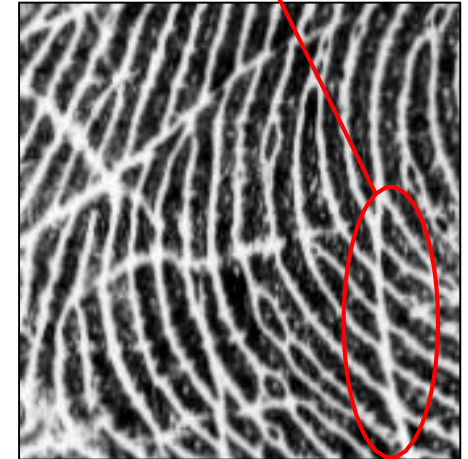
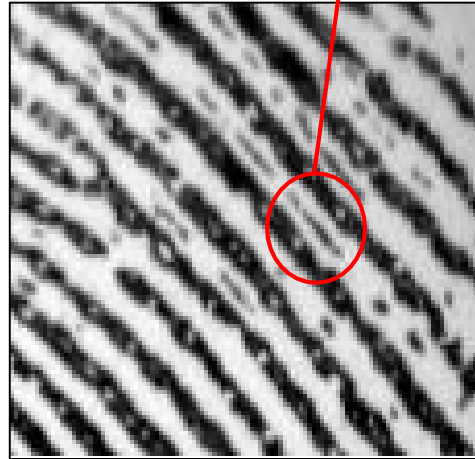


frequency
image

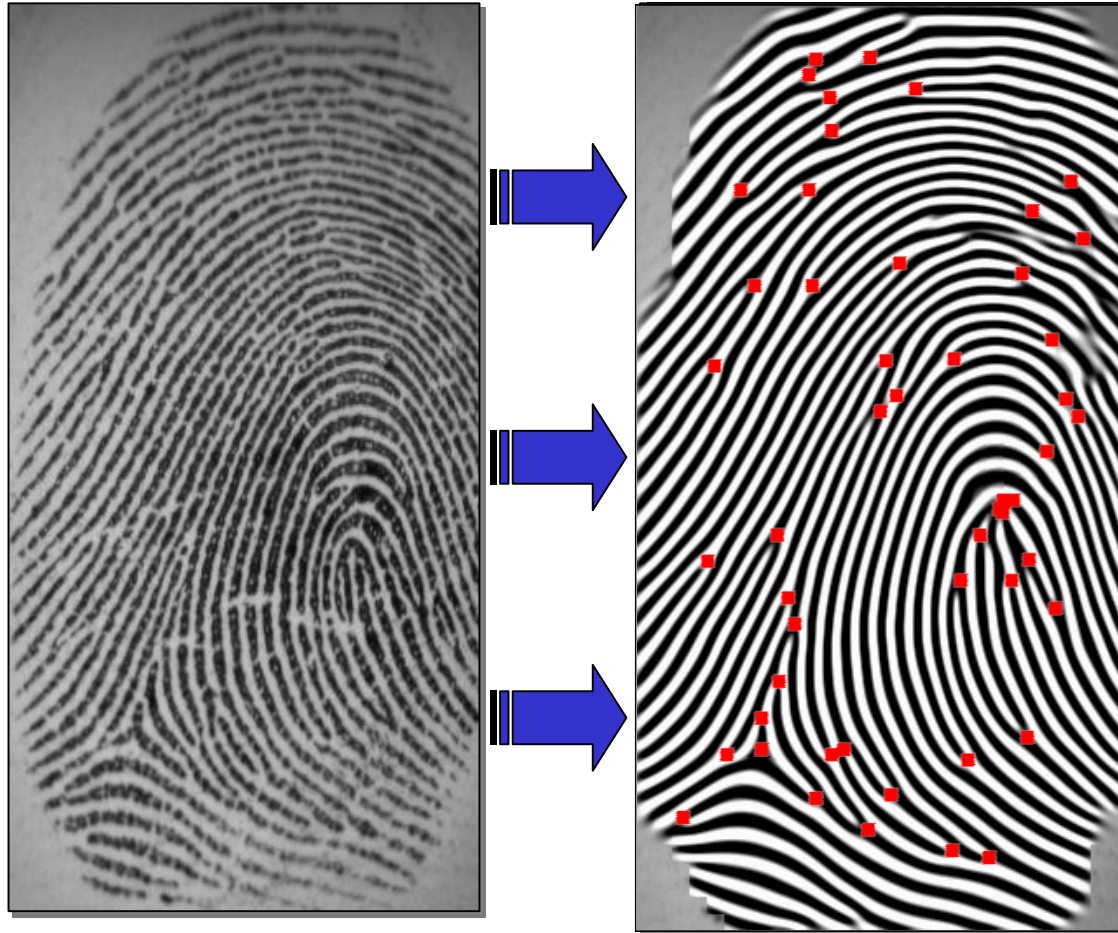
- The ridge-line flow can be effectively described by an **orientation image**
- The ridge line density can be summarized using a **frequency image**

Fingerprint Anatomy

- At the very local level (e.g., acquisition at 1000 dpi) it is possible to identify sweat pores (from 60 to 250 μm), incipient ridges, creases, etc.

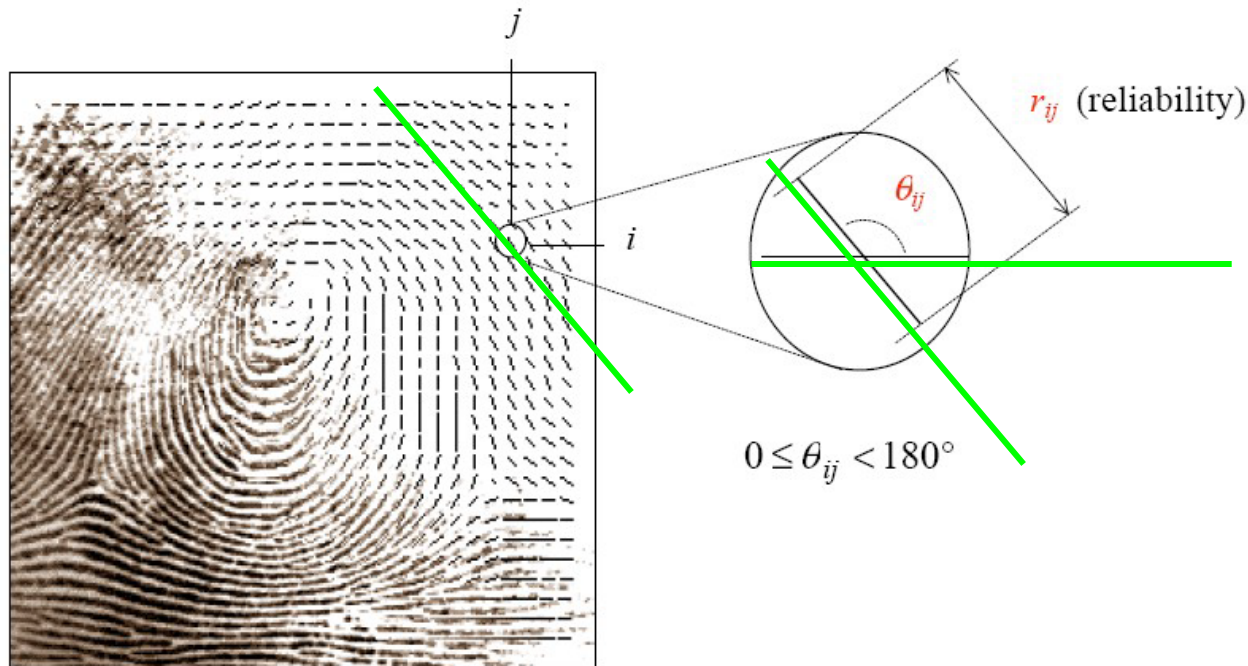


Feature Extraction Steps



Local ridge orientation

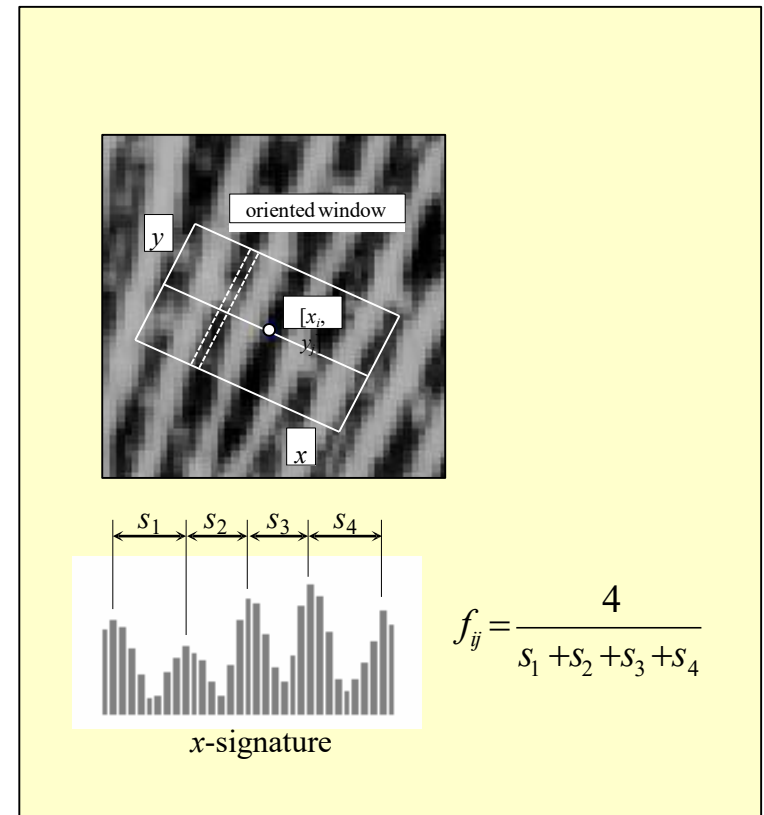
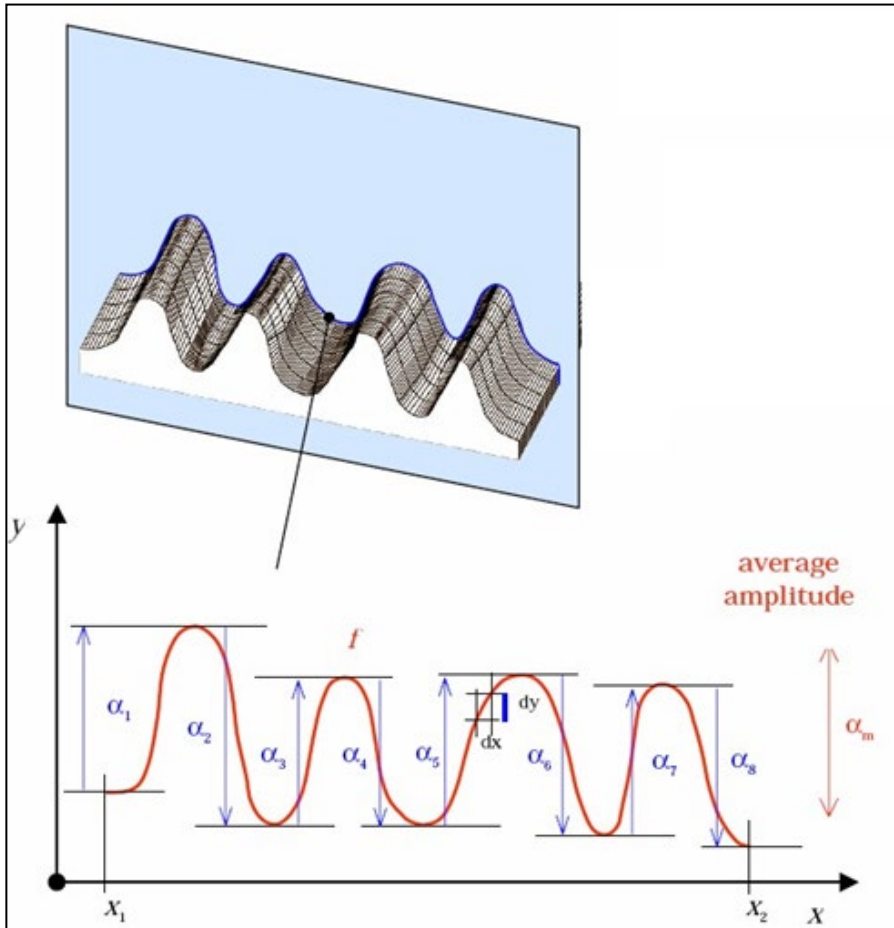
The local ridge orientation at $[x,y]$ is the angle $\theta_{xy} \in [0..180^\circ[$ that the fingerprint ridges, crossing through an arbitrary small neighborhood centered at $[x,y]$, form with the horizontal axis.



The simplest and most natural approach for extracting local ridge orientation is based on computation of **gradient phase angles** (problems of **non-linearity** and **circularity**).

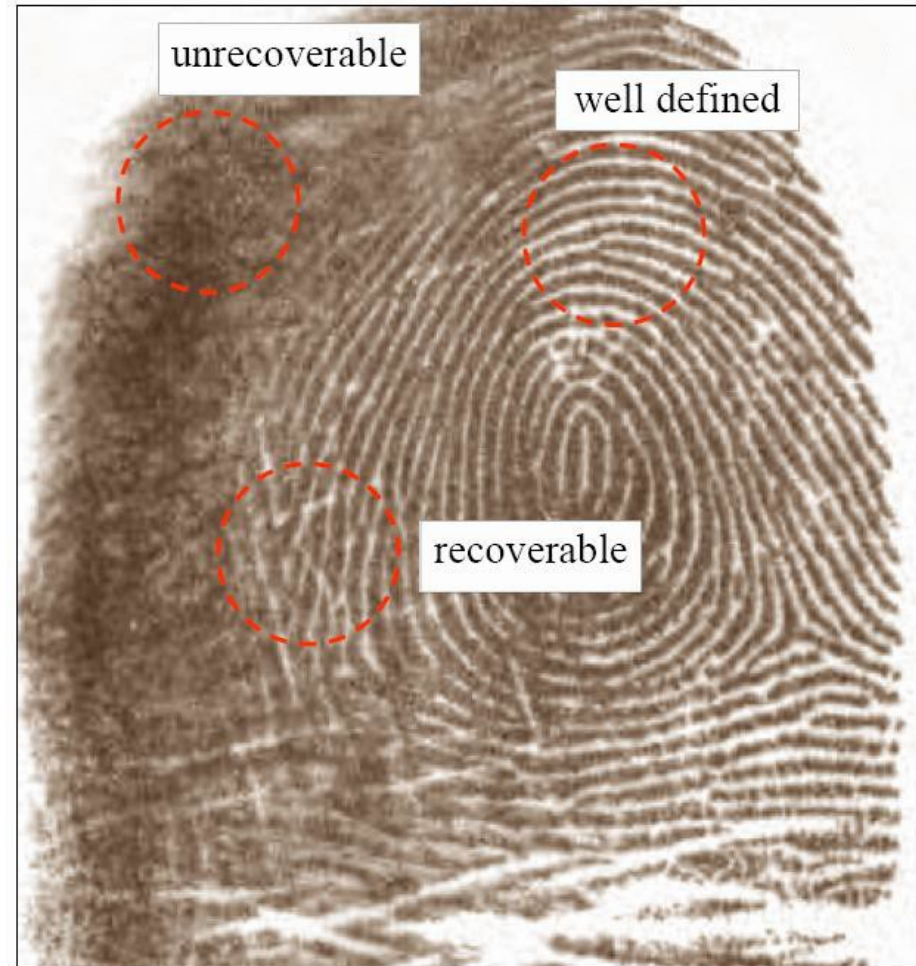
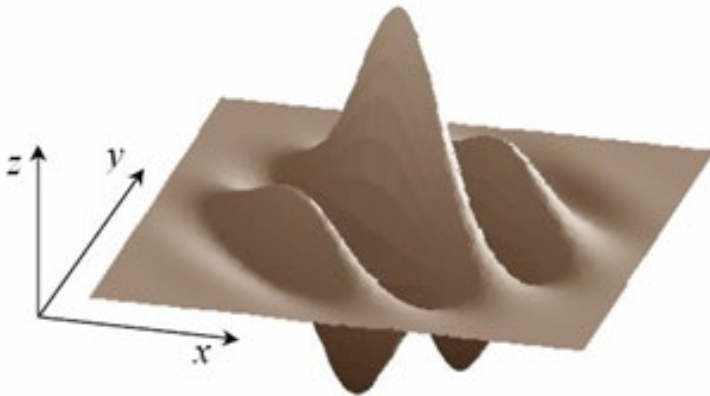
Local ridge frequency

- Number of ridges per unit length along a hypothetical segment orthogonal to the local ridge orientation

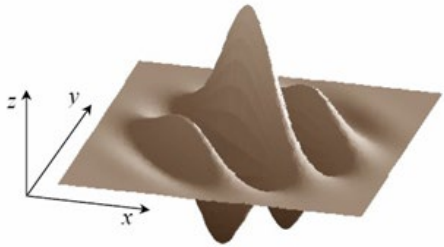


Local enhancement

- Aimed at **improving the quality** of **recoverable regions** to simplify successive stages
- Uses **orientation information** obtained previously
- Apply a **filter** with "looks like" the ideal pattern (E.g. Gabor filter)

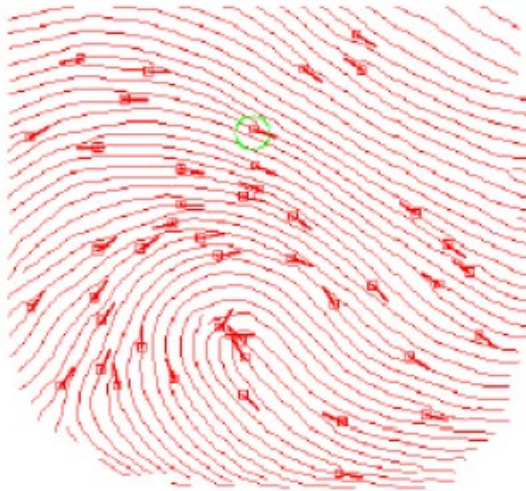


Local enhancement

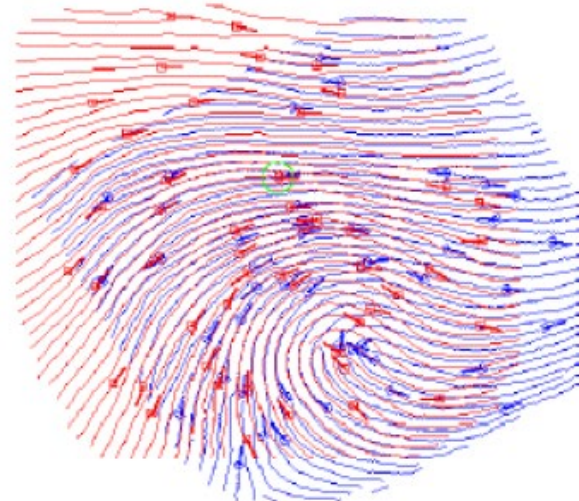


Fingerprint Matching

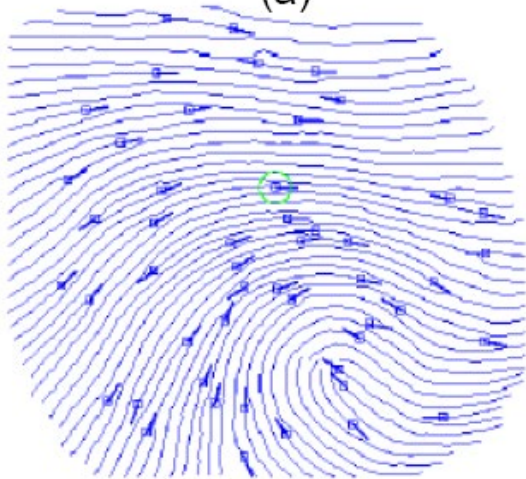
Finding the alignment that results in the maximum number of minutiae pairings.



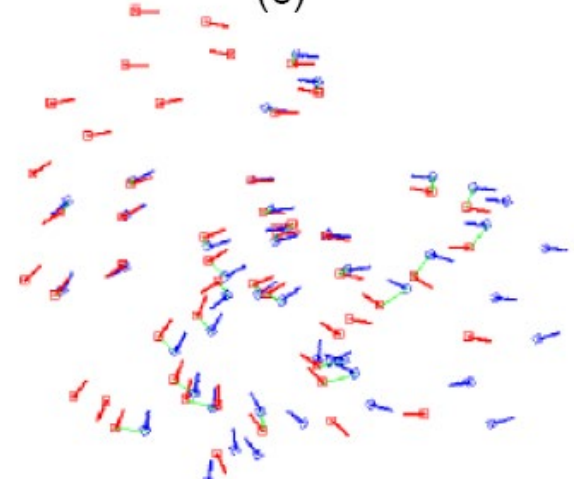
(a)



(c)



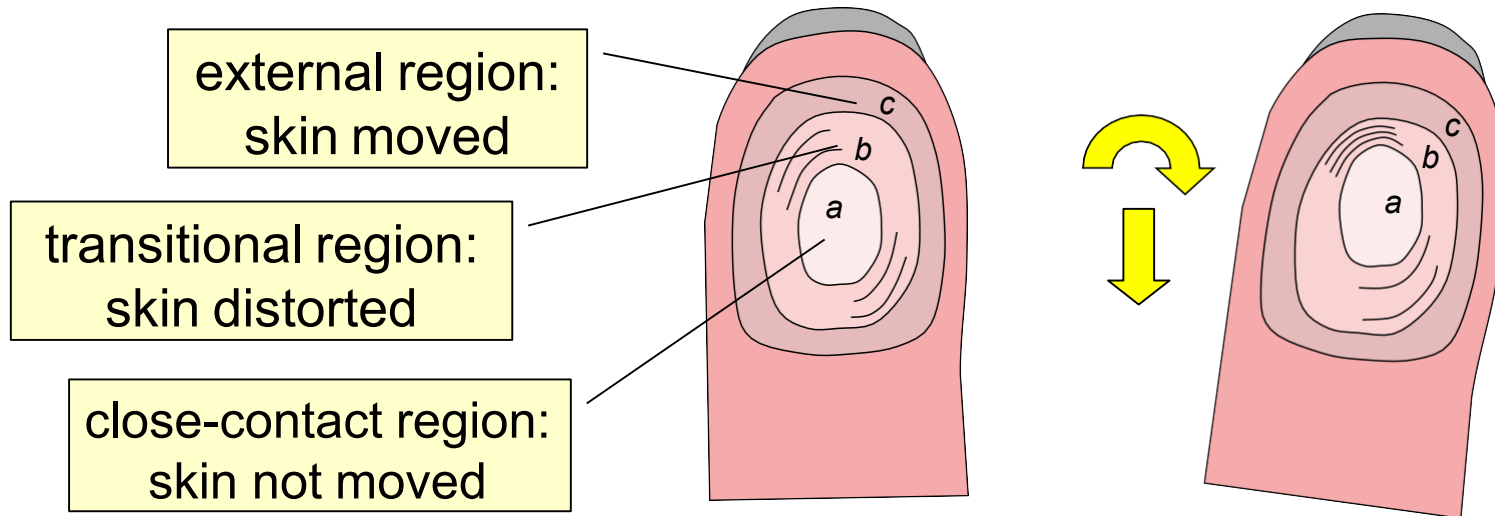
(b)



(d)

Skin distortion

The finger pressure against the sensor is not uniform, but decreases moving from the center towards the borders.

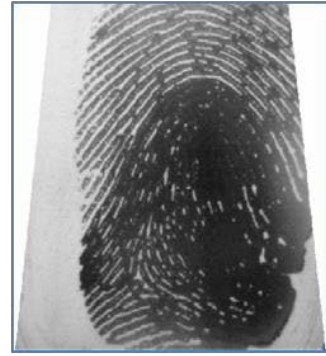


- a) close-contact region, where the high pressure and the surface friction does not allow any skin slippage
- c) external region, where the low pressure allows the skin to be dragged by the finger movement
- b) transitional region where an elastic distortion is produced to smoothly combine regions a and c

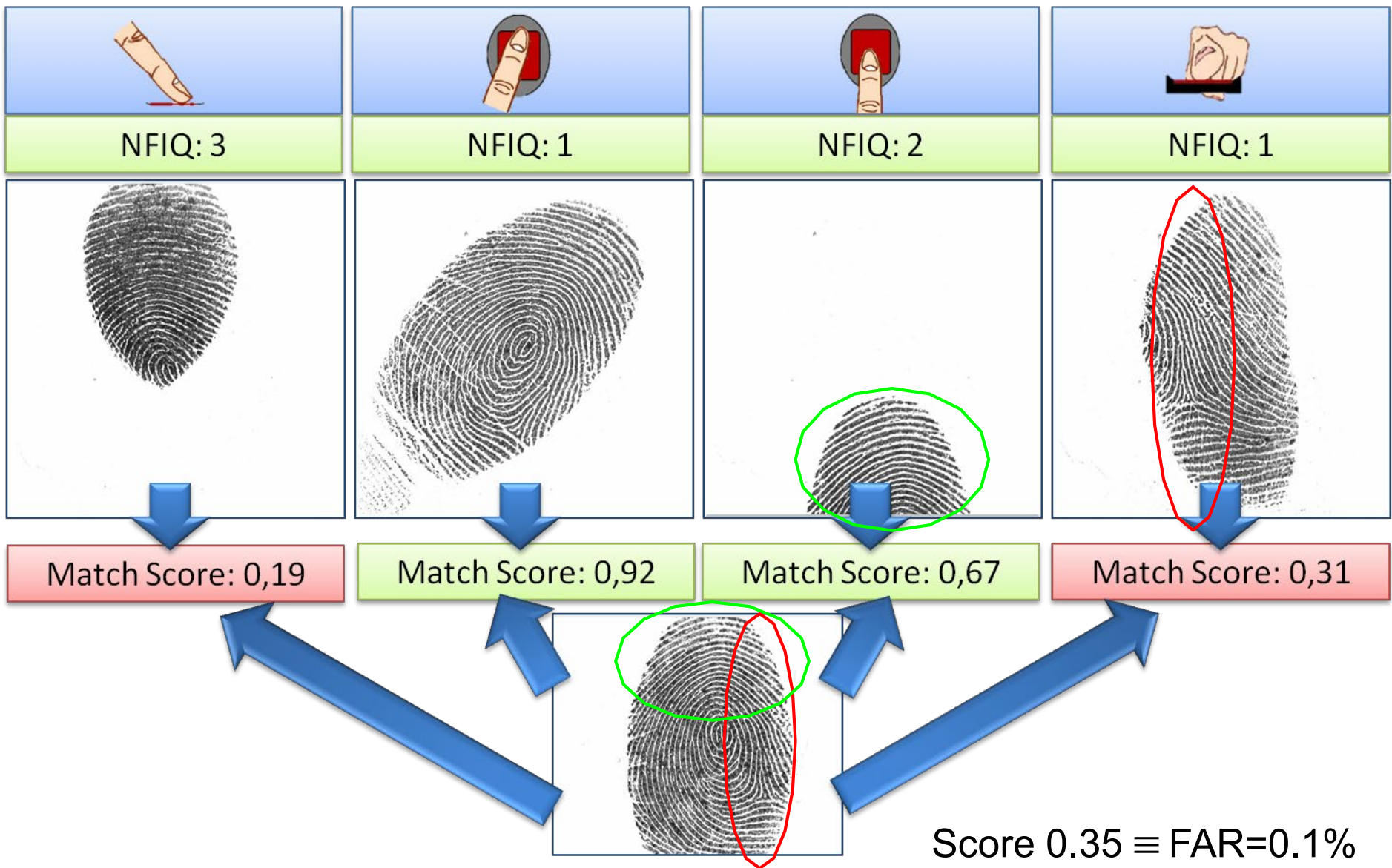
Fingerprint: main challenges

- High **displacement** and/or **rotation**
 - Small overlap between the template and the input fingerprints. Particularly serious for *small-area sensors*.
 - A finger displacement of 2 mm (imperceptible to the user) results in a translation of about 40 pixels in a fingerprint image scanned at 500 dpi.
- Non-linear **distortion**
 - The three-dimensional shape of a finger has to be mapped onto the two-dimensional surface of the sensor.
 - This results in a non-linear distortion in successive acquisitions of the same finger due to skin plasticity.
- Different **pressure** and **skin** condition
 - Non uniform finger pressure, dryness of the skin, skin disease, sweat, dirt, grease, and humidity in the air.
- Feature extraction **errors**
 - Feature extraction algorithms are imperfect and often introduce measurement errors, in particular in *low-quality fingerprint images*

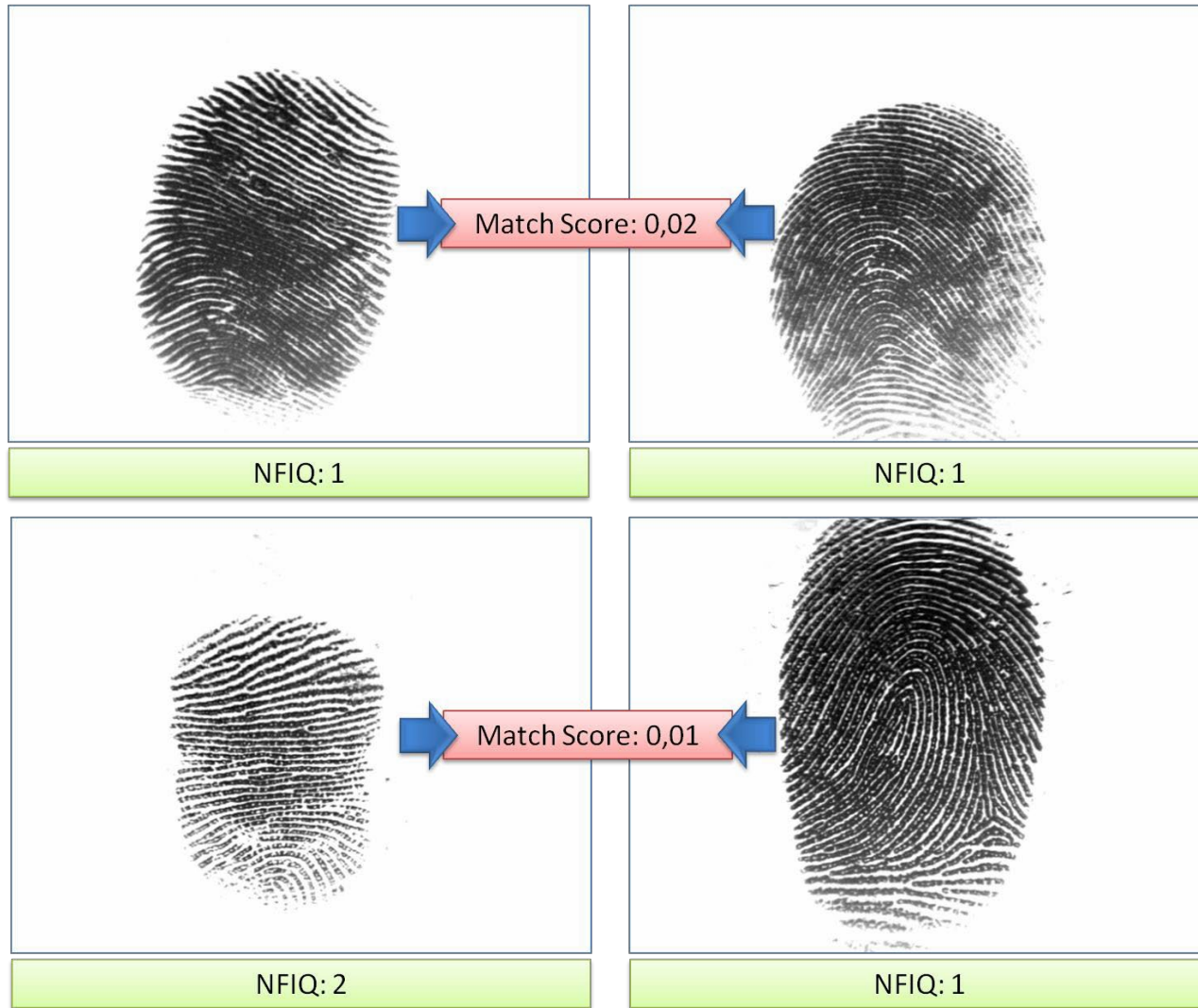
Fingerprint: main challenges



Displacement or rotation

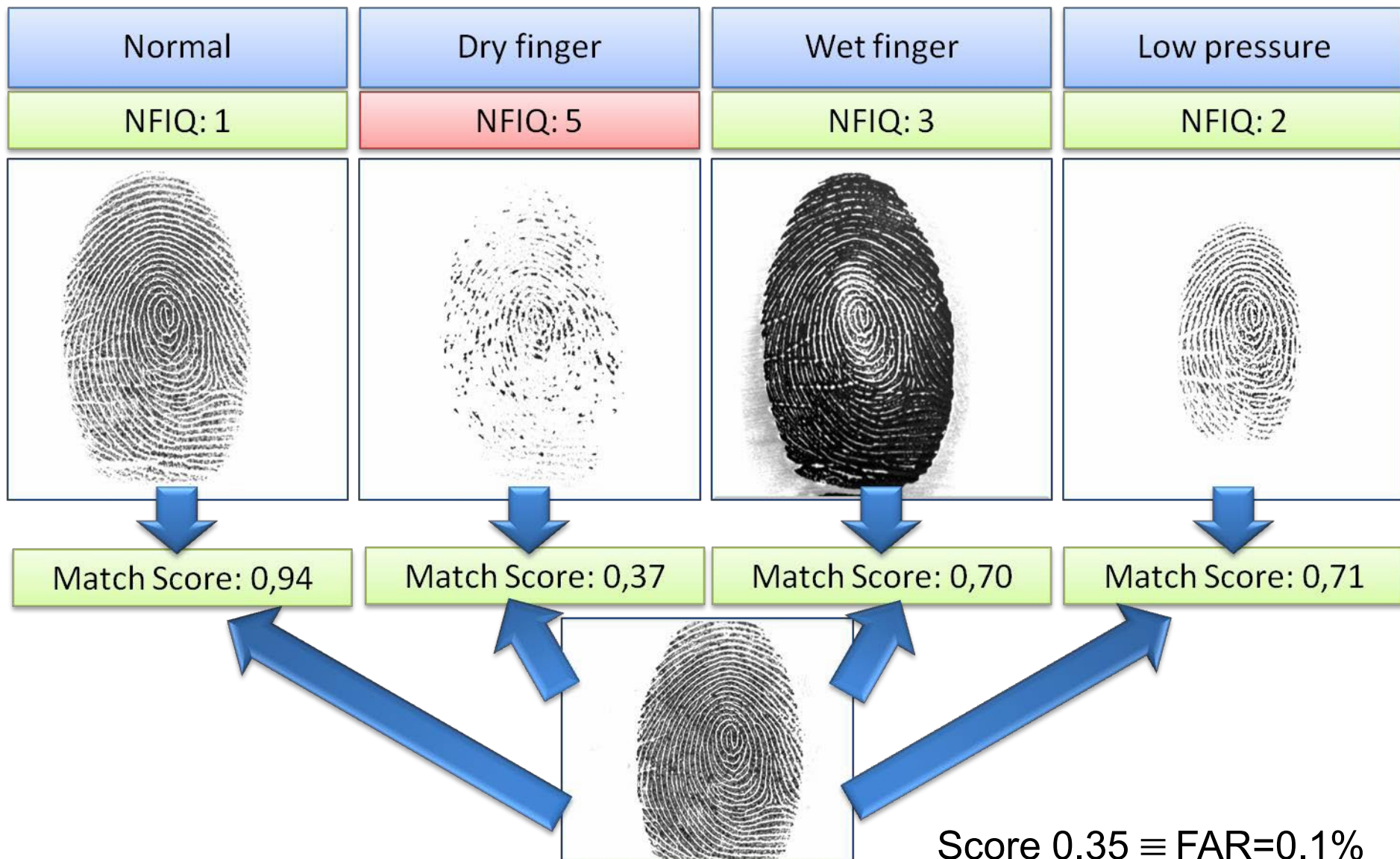


Non-linear distortions



Score 0.35 $\equiv$ FAR=0.1%

Different pressure and skin condition



Feature Extraction Errors

Automatic minutiae extraction is difficult in low quality images

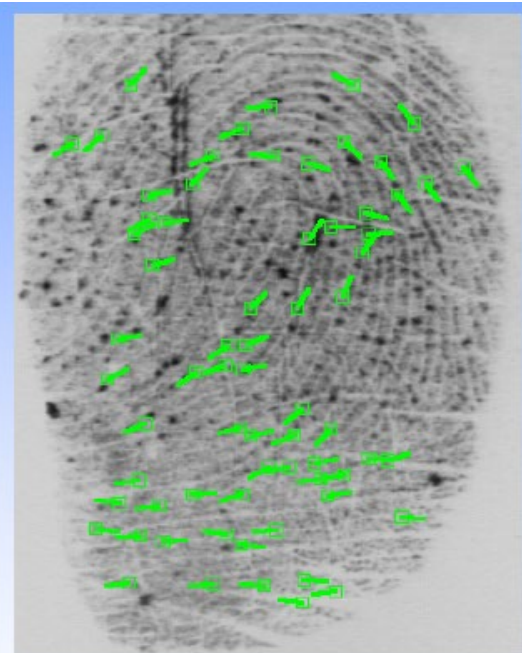
- True minutiae are missed
- False minutiae are introduced



Quality Index = 0.96
False Minutiae = 0



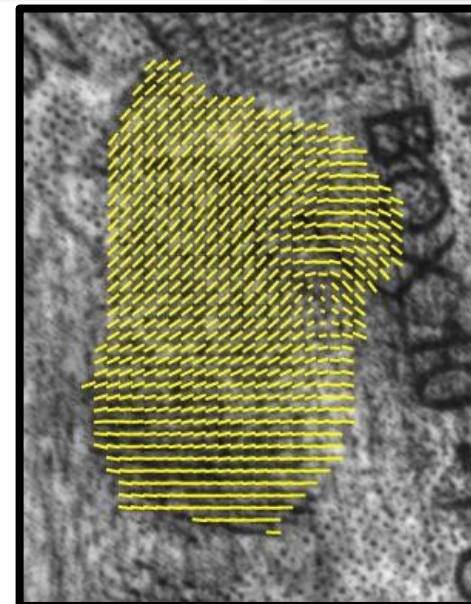
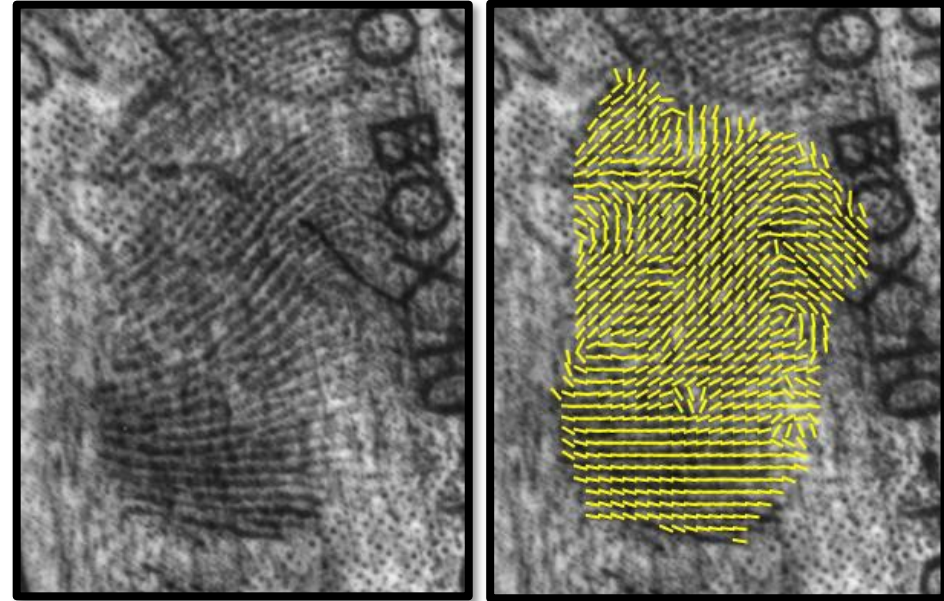
Quality Index = 0.53
False Minutiae = 7



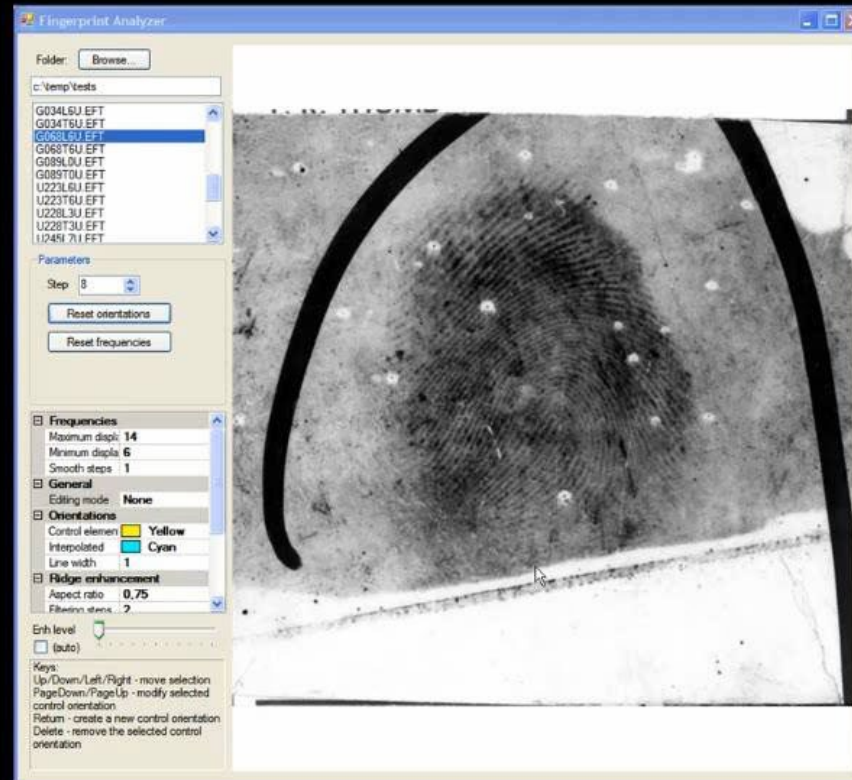
Quality Index = 0.04
False Minutiae = 27

Enhancement of latent fingerprints

- A very challenging task
- How to (automatically) estimate the context?
 - Can local orientations and frequencies be reliably computed on very low-quality fingerprints?
 - The problem is still open.

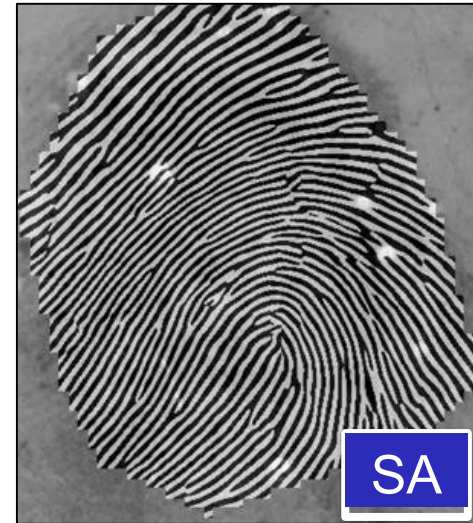
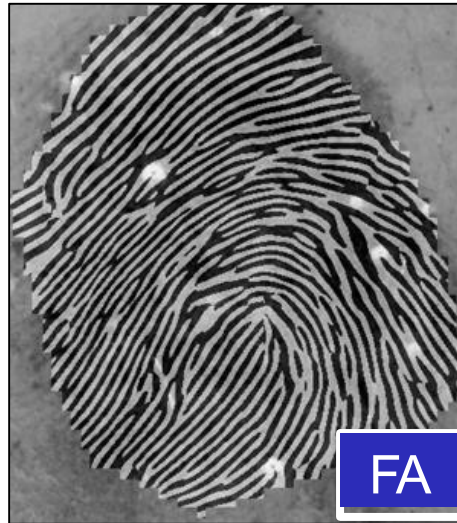
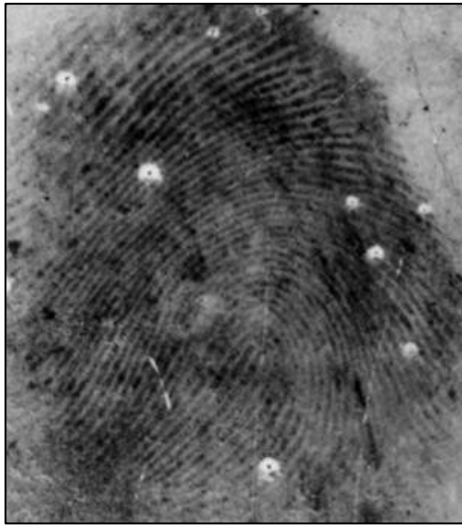


A semi-automatic approach



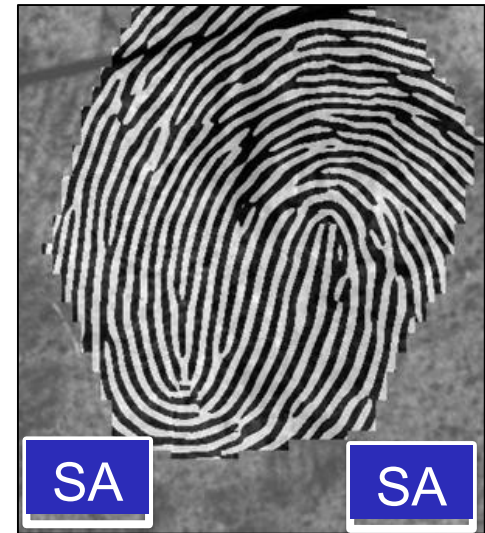
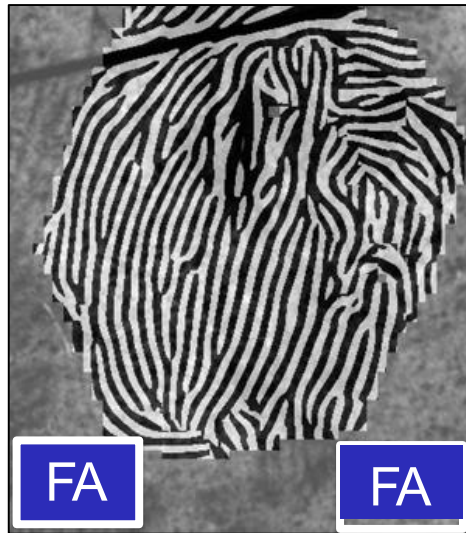
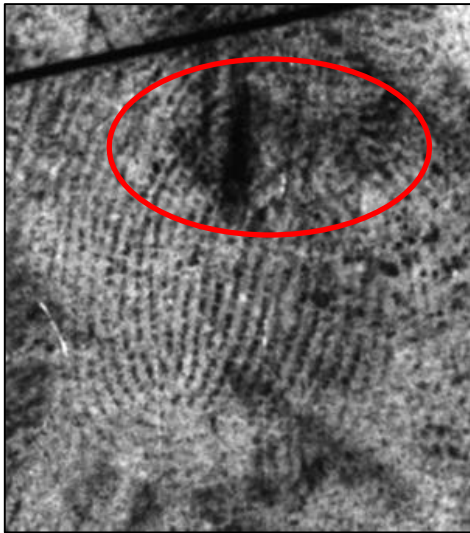
A semi-automatic approach

Example: a NIST DB27 “good” latent image



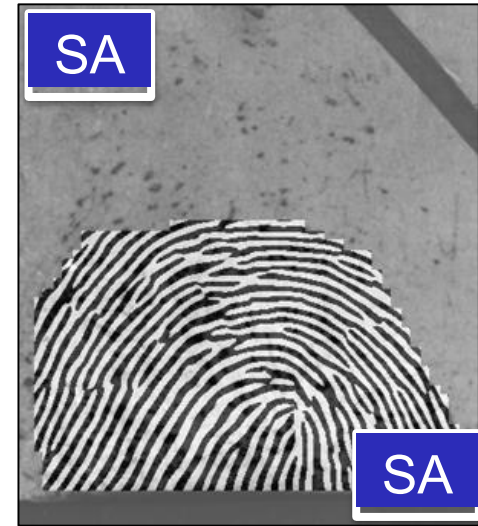
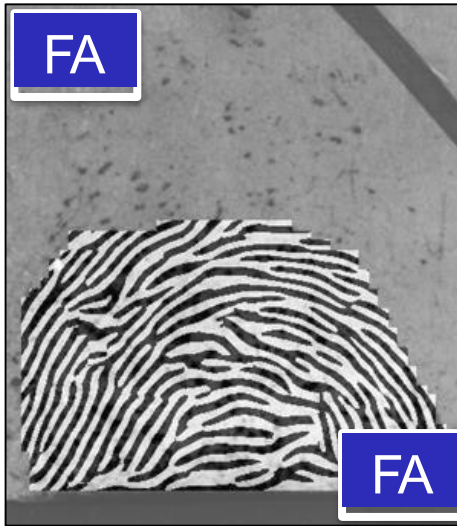
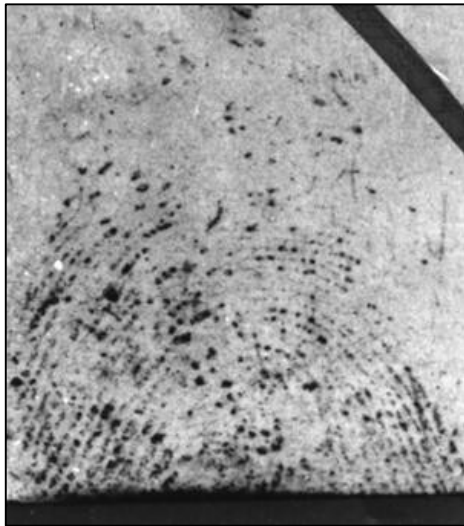
A semi-automatic approach

Example: a NIST DB27 “bad” latent image



A semi-automatic approach

Example: a NIST DB27 “ugly” latent image





Fingerprints: Open Directions

- Nowadays research on fingerprints is active on:
 - Semi-automated and Automated Latent processing and matching
 - Fast fingerprint matching and indexing (millions of fingerprints)
 - New very large AFIS: Indian UIDAI (enrolling 1.2 billions residents), European BMS (for the new Visa Information System)
 - Increasing demand for low cost AFIS from emerging countries
 - New scanning techniques
 - Fingerphoto recognition
 - Embedding transparent fingerprint sensors in the smartphone glass
 - Fake fingerprint detection & template protection techniques

Outline

Overview of Main Biometric Modalities

- Fingerprint
- **Face**
- Iris

Other Biometric Modalities

Why Face?

Face recognition has several advantages over other biometric modalities such as fingerprint and iris

- **Most common human experience**
- **Wide availability** and **use of cameras and databases**: passport, driver license, research-based, Youtube...
- **Easy to capture**: at a *distance* and in a *non-cooperative manner*

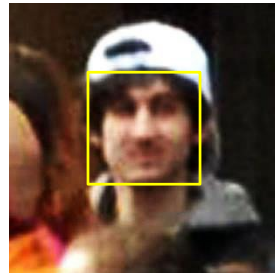
Variety of Cameras



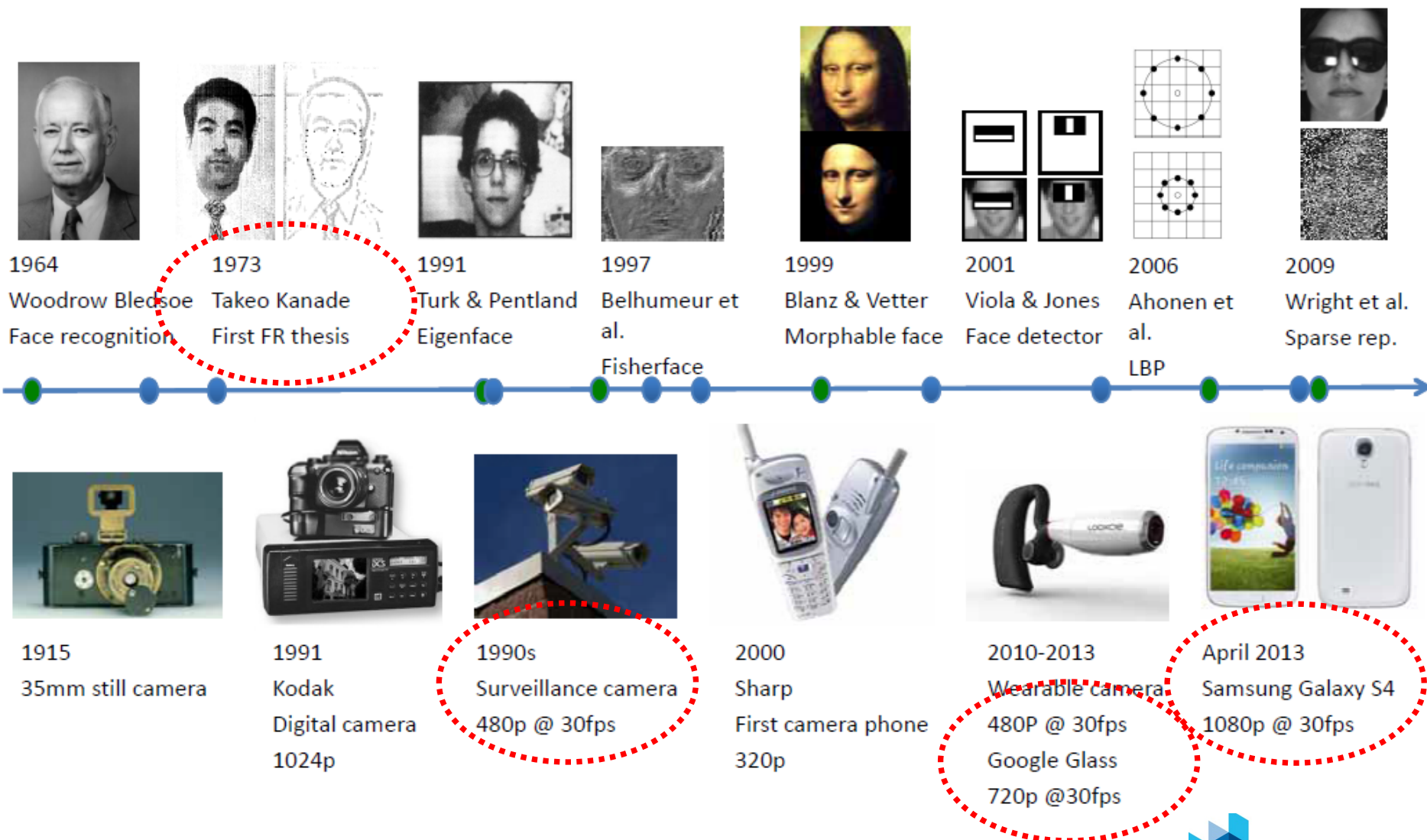
Face Image Technology in the Forefront

Face recognition has become increasingly important due to

- Rapid advances in image capture devices (surveillance cameras, mobile phone cameras)
- Availability of huge amounts of face images on the Web
- Increased *demands* for higher security



Face Recognition Milestones



Why FR is Difficult?

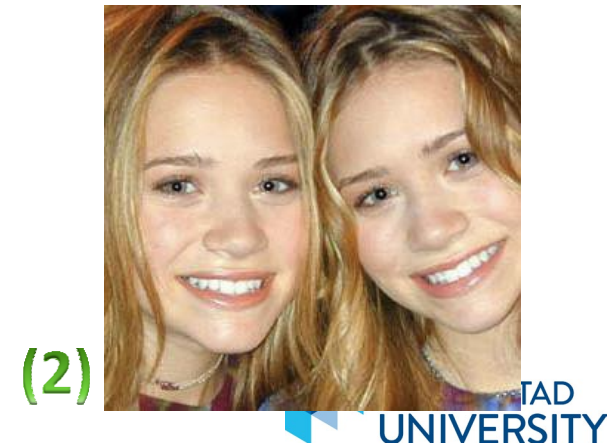
- Intra-class variability (1)
- Inter-class similarity (2)

- Uncontrolled Conditions

- *Aging*
- *Pose*
- *Illumination*
- *Expression*
- *Distance*
- *Motion*



- Segmentation (finding the face) (1)
- Scalability
- Privacy Issues



Illumination and Pose Variations

Controlled Conditions

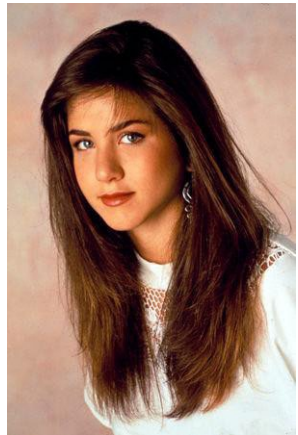


Uncontrolled Conditions



Aging (intra-class variability)

1993



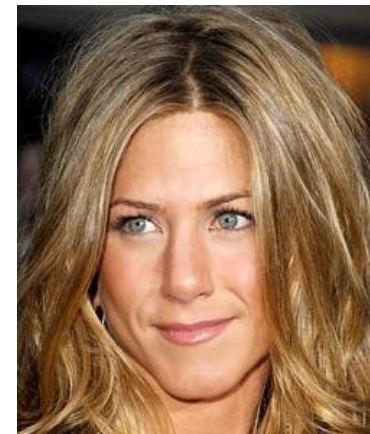
1999



2004



2008



2011



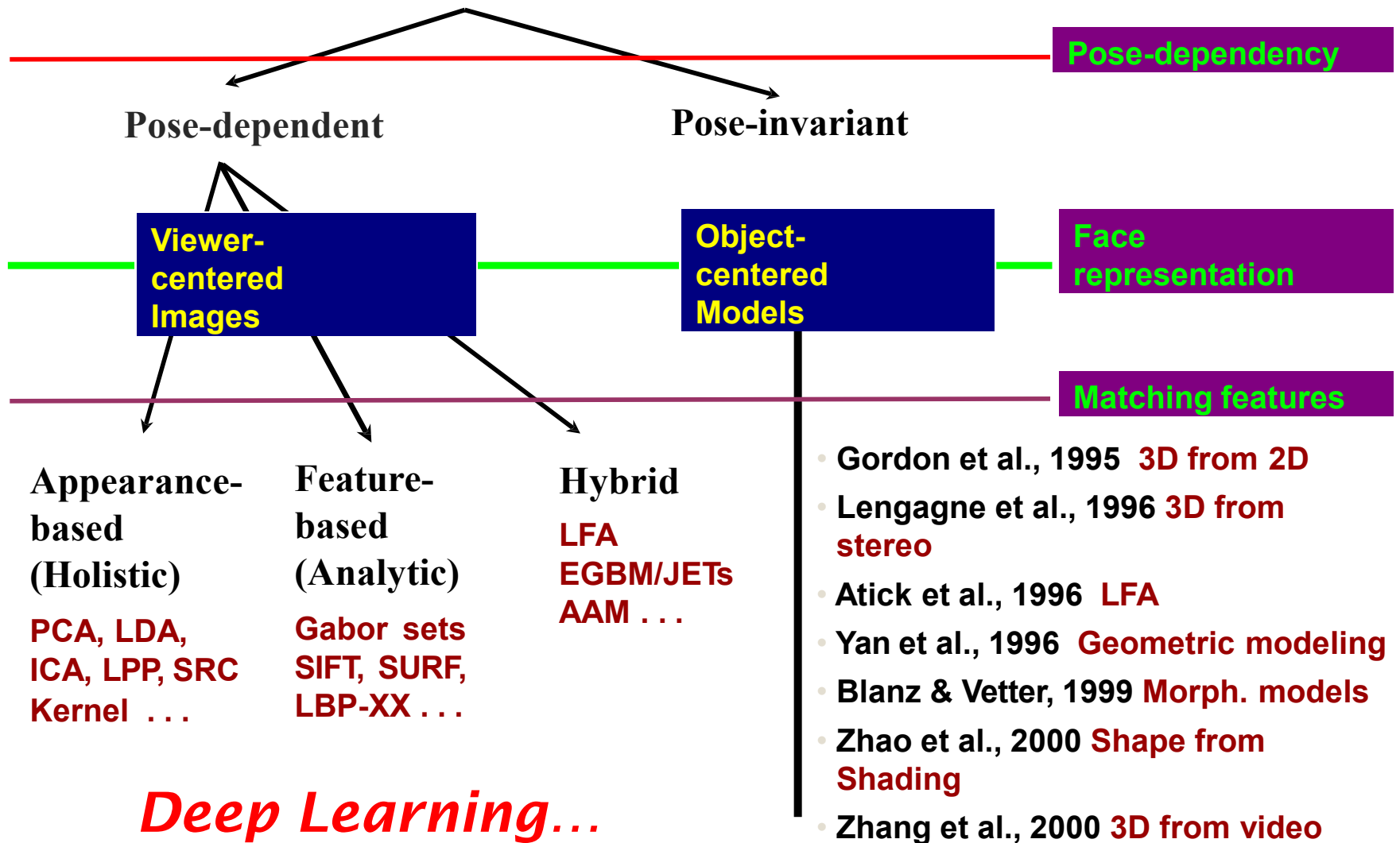
Surgery (intra-class variability)



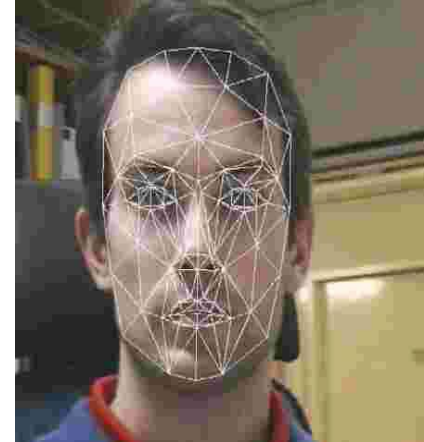
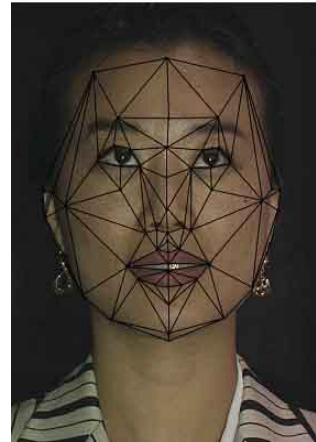
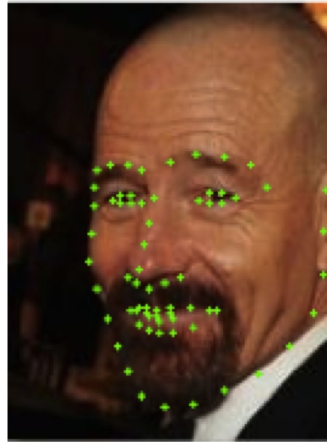
Makeup (intra-class variability)



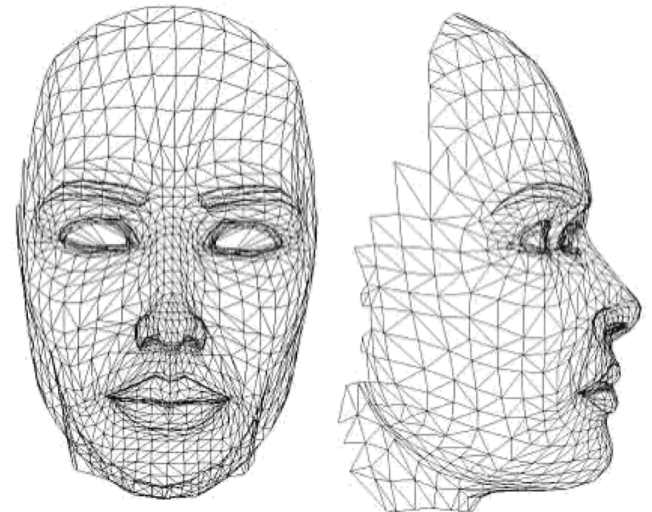
Taxonomy of Methods for Face Recognition



Local (Landmarks) Face Recognition



- 2D landmarks can be defined and tracked on face images
- Simple 2D vs complex 3D representations
- Not easy to detect in unconstrained conditions

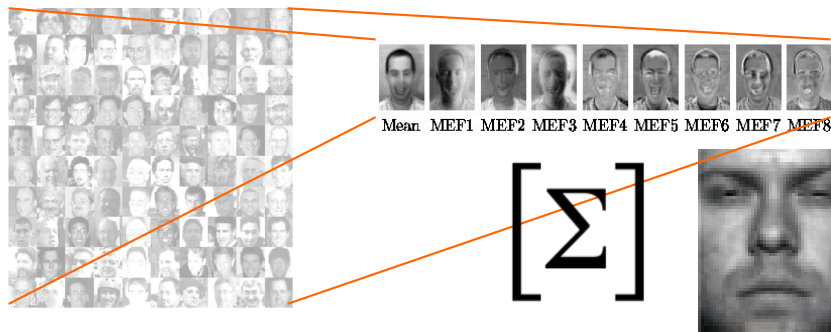


Holistic Face Recognition

PCA



LDA



$[\Sigma]$



$= \sigma_1$



$+ \sigma_2$

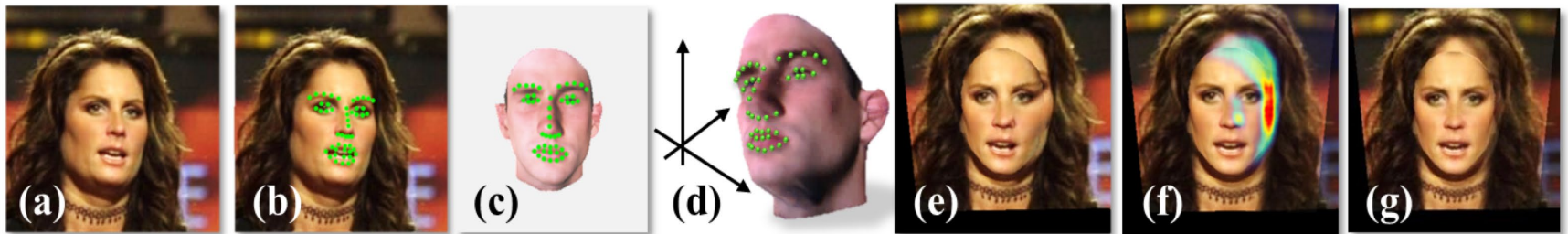


$+ \dots + \sigma_n$



- Represent any given face as a combination of a set of "universal" basis faces
- The set of basis faces is learnt statistically from a training database

Dealing with pose: 2D frontalization



- (a) Query photo; (b) landmarks detection; (c) textured 3D computer graphics model with landmarks; (e) The estimated projection matrix is used to back-project the query intensities to the reference coordinate system; (f) Estimated visibility overlaid on the frontalized result. Warmer colors reflect less visible pixels. (g) Facial appearance produced by borrowing texture from symmetric face areas.



Working environments



Working environments

Sketch to photo comparison

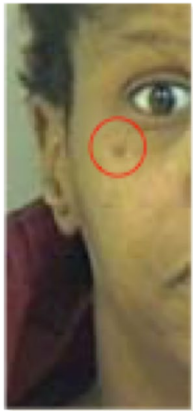


The search for the Unabomber generated one of the most well known composite sketches in recent history. The sketch is based on a single 1987 witness, who saw an individual drop off a package containing a bomb. Although the identity is obscured by a hooded sweatshirt and dark glasses, there are some very obvious differences between the sketch and Ted Kaczynski, the man who was later arrested for the crimes. The man in the sketch seems to have a very narrow nose, yet Kaczynski has a prominent, almost bulbous nose. Kaczynski also has a broad chin and prominent age lines extending from above the flare of his nose down. The basic shape of the face is different in the two sketches, but neither accurately captures Kaczynski's likeness.

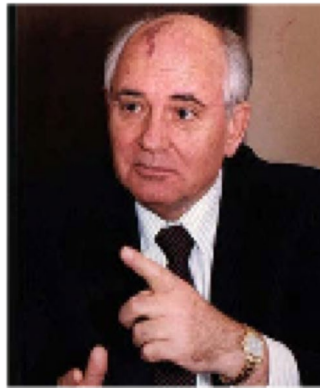
CCTVs



Facial Marks



Mole in partial view



Birth mark



Mole in side view



Face tattoo

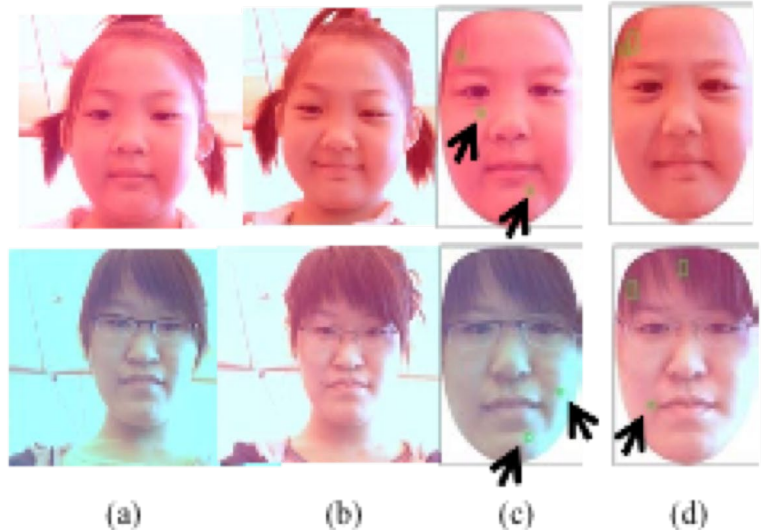
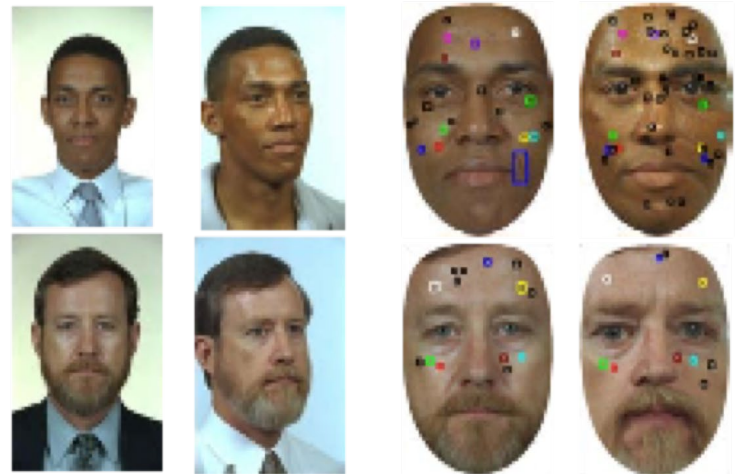


Figure 10. Example image pairs where facial marks helped to improve the matching performance. (a) (b) Probe and gallery images for the first two rows and two images of identical twins for the third and fourth rows. (c) (d) Facial mark detection results from (a) and (b).



Face: Open Directions

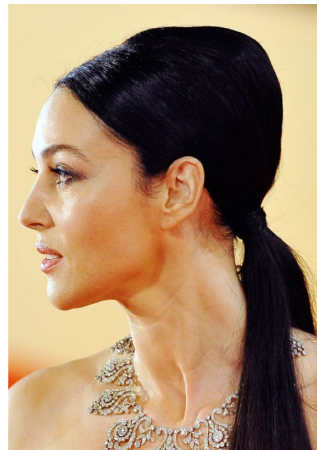
- Nowadays research on face is active on:
 - Age - Pose - Illumination – Expression (APIE)
 - APIE-invariant representations
 - Subject-specific representations
 - Mobile Applications
 - Fake face detection, spoofing & template protection techniques

Four Big Problems

A – Aging



P – Pose



I – Illumination

E - Expression

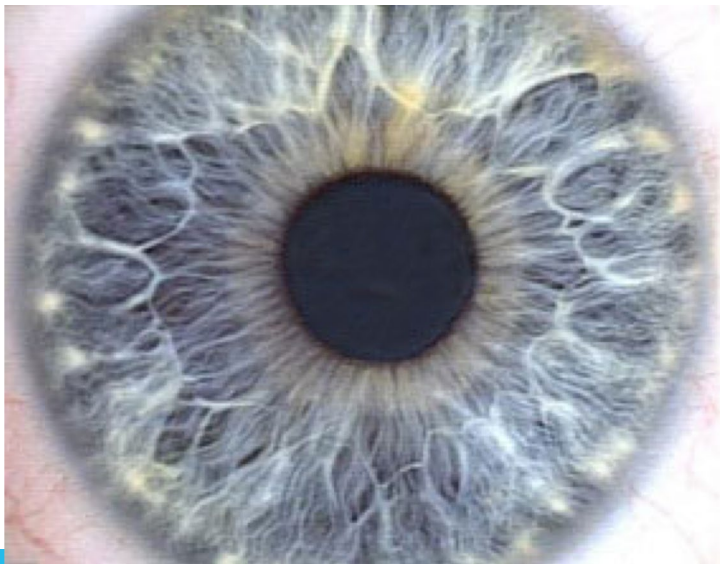
Outline

Overview of Main Biometric Modalities

- Fingerprint
- Face
- Iris

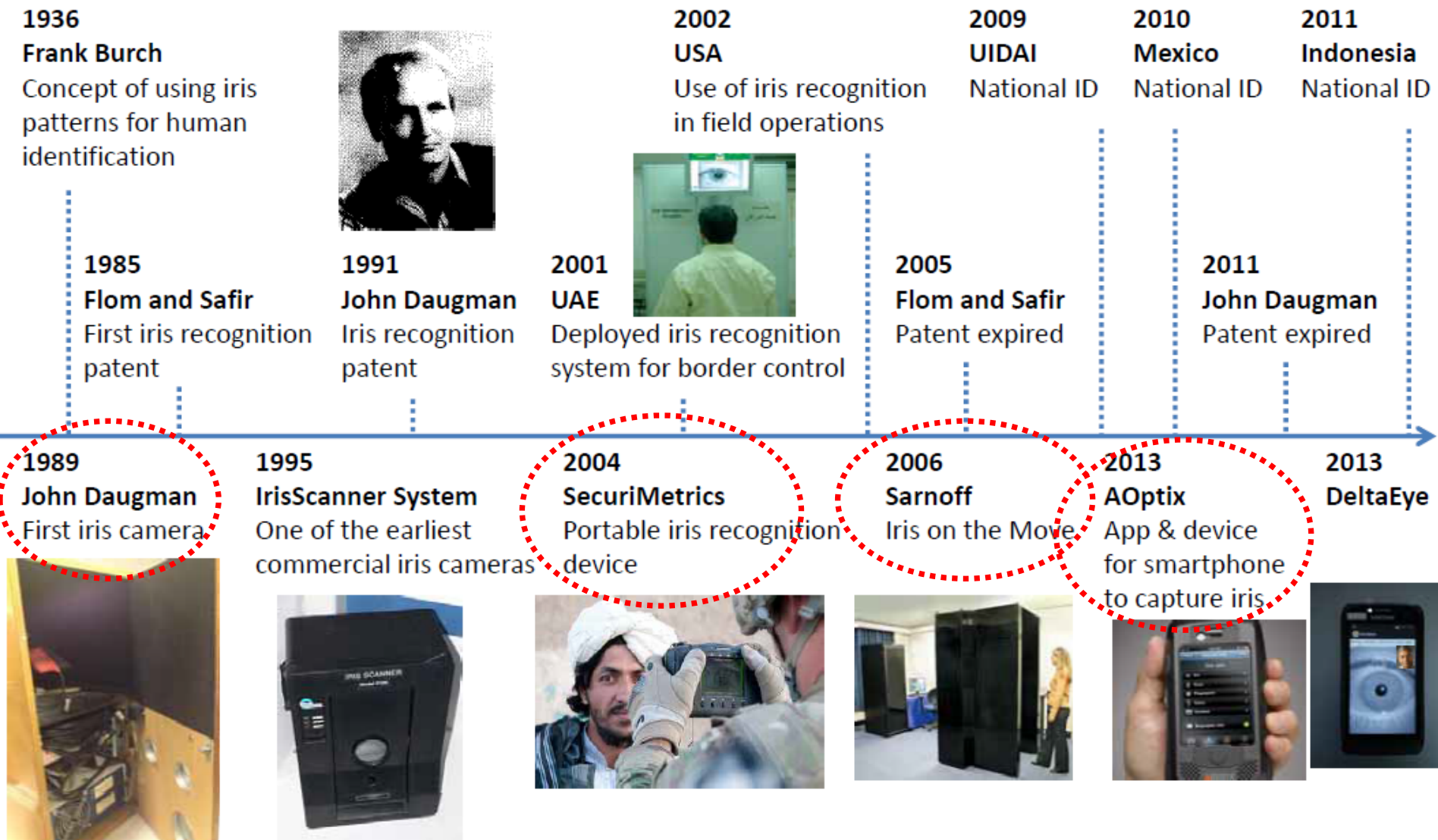
Other Biometric Modalities

Color Iris



- Iris is the **annular region** of the eye bounded by the pupil and the sclera
- This (very rich) visual **texture** of the iris stabilizes during the first two years of life and carries distinctive information useful for recognition
- Each iris is believed to be **unique**; even irises of **identical twins** have been observed to be different

Iris Recognition Milestones



Impact of Light

- The primary function of the iris is to **regulate the amount of light** entering the eye by dilating or contracting a small opening in it called the **pupil**
- The iris **contracts** the pupil when the ambient illumination is **high** and **dilates** it when the illumination is **low**

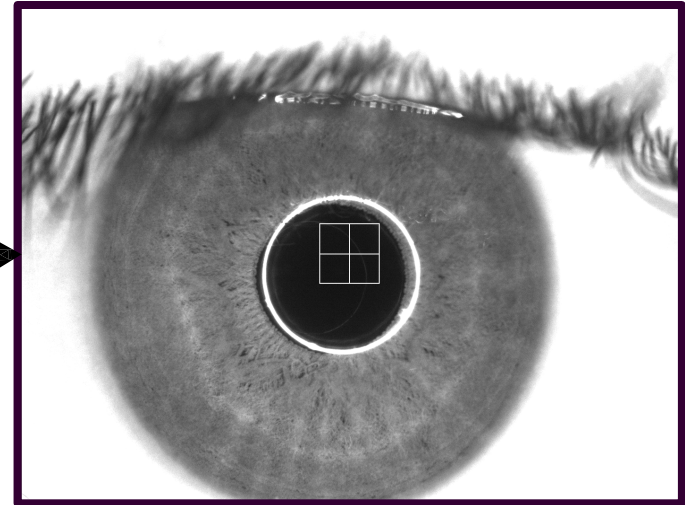
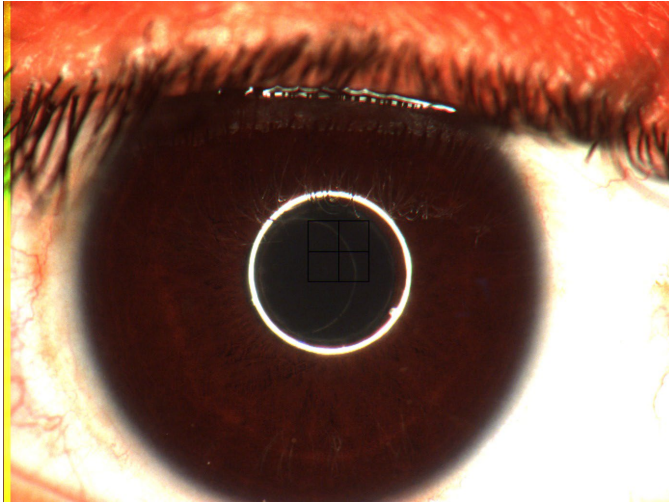


Near-Infrared illumination (NIR)

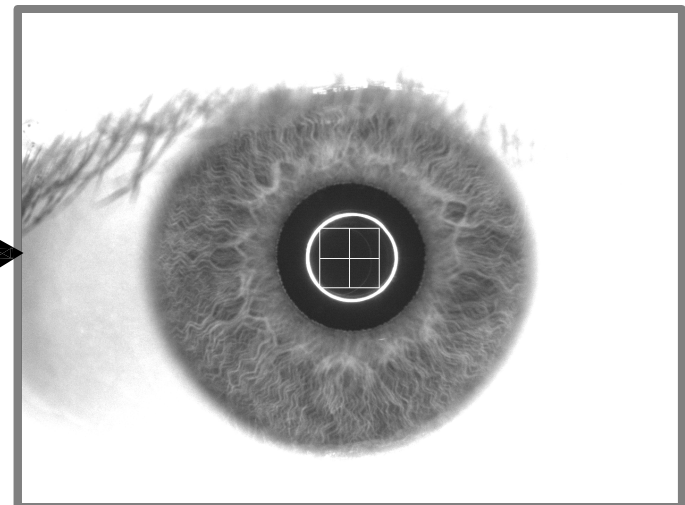
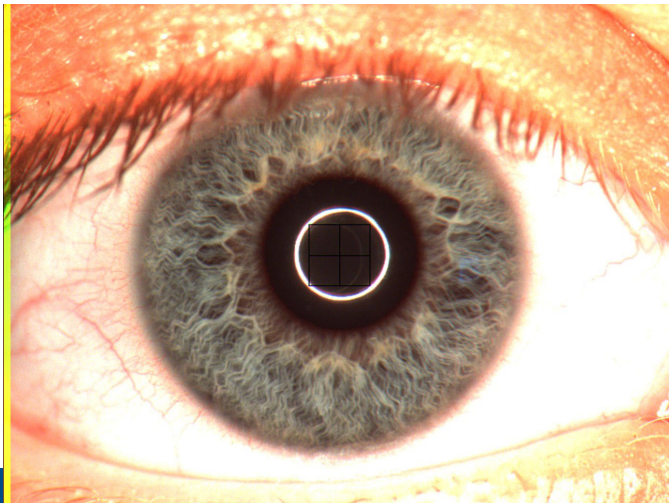
COLOR IMAGE

NIR IMAGE

DARK IRIS



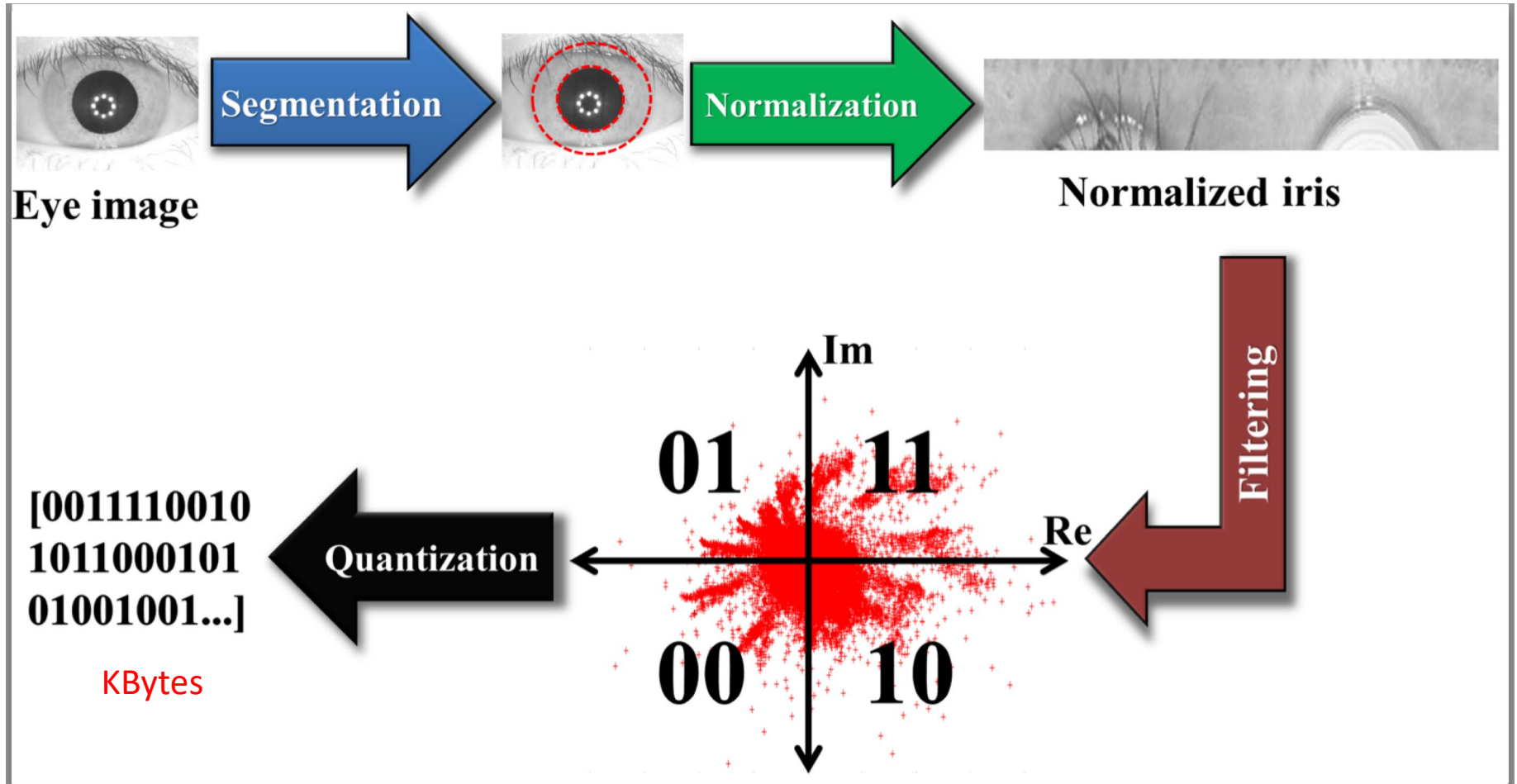
LIGHT IRIS



Why NIR?

- **Dark-colored Iris:** The textural details of dark- colored irides (majority of the world population) are more evident in the NIR channel than in the red, green, or blue channels
- **Non-intrusive:** NIR light cannot be perceived by the human eye. This ensures that the image acquisition process is non-intrusive, even when the eye is required to be in close proximity to the sensor and the NIR light source.
- **Pupil dilation:** the effect is greatly eliminated by NIR light

Iris Recognition Modules



Iris Segmentation

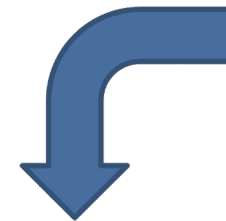
- Assumptions

- Iris boundaries are approx. circular
- Iris boundaries are stronger than any other circle in the image

- Method

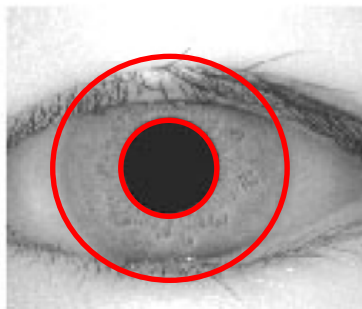
- Filters tailored to detect circles

circular filter

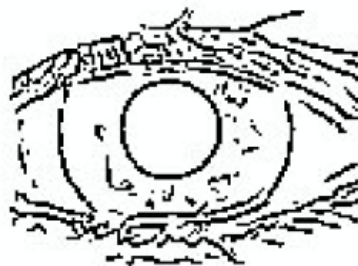


1st detect inner circle
no disturbance
of the other

then detect outer circle
outside the
inner circle



original



edge map

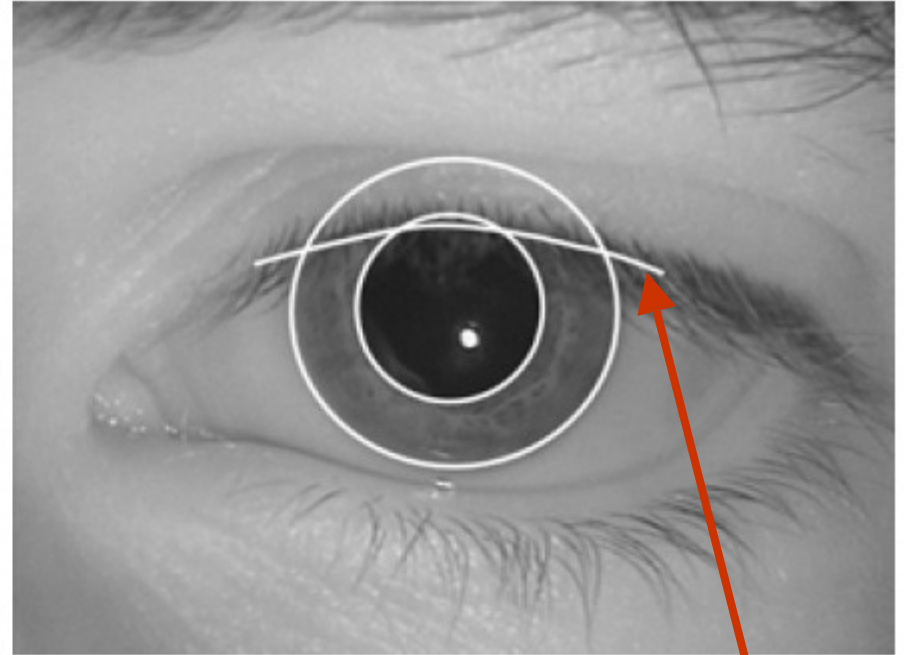
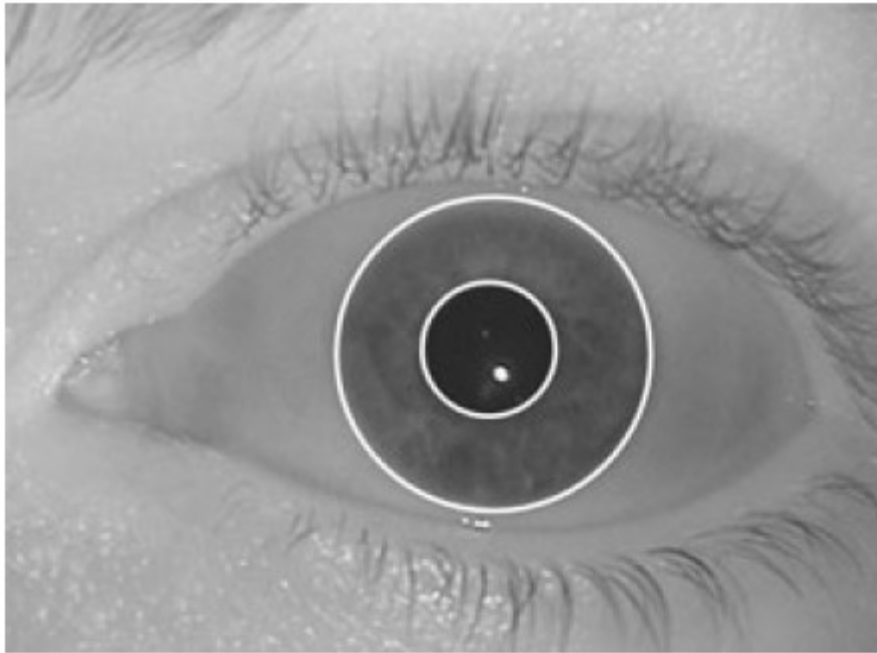


horizontal
edge map



vertical
edge map

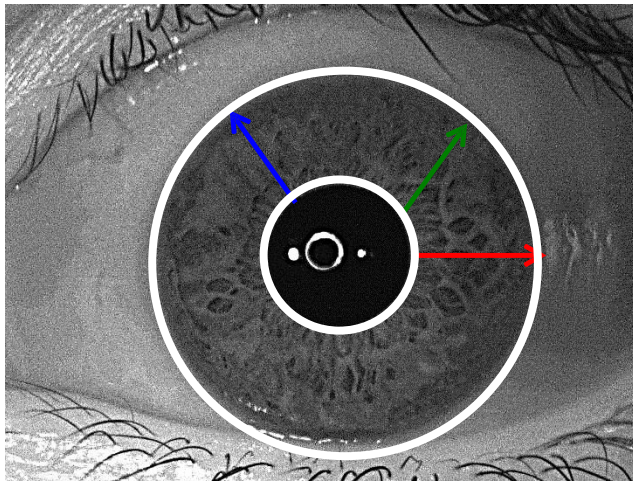
Example Iris Segmentation



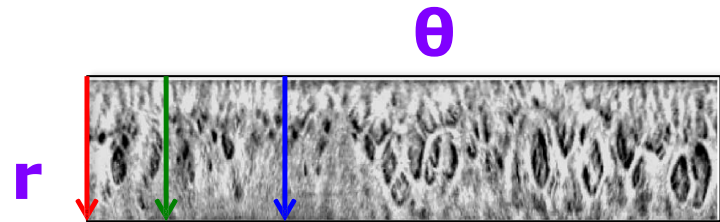
**Eyebrow found through
post-processing**

Iris Normalization

- **Challenge with iris**
 - Size of iris and pupil varies (dilation/contraction due to light, distance to camera)
- **Solution**
 - “Normalize” the iris by unwrapping it into a rectangular region



Cartesian Coordinates



Pseudo Polar Coordinates

Iris Sensors



(a)



(b)



(c)



(d)



(e)



(f)



(g)



(h)



(i)

- (a) LG IrisAccess
- (b) Panasonic BM-ET
- (c) Oki IrisPass-M series

- (d) Oki IrisPass-H series
- (e) DataStrip Easy Verify
- (f) Eyelock Myris

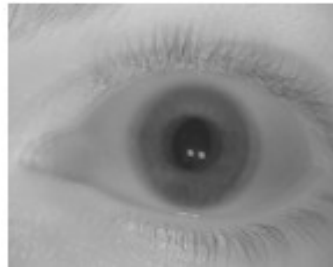
- (g) Hoyos Hbox
- (h) Sarnoff Iris On the Move
- (i) AOptix Insight

Factors negatively affecting iris images

occlusion



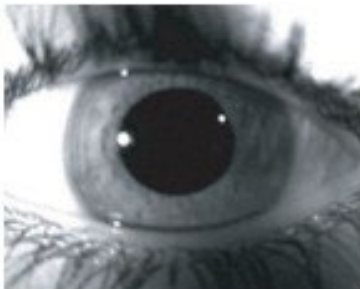
motion blur



defocus



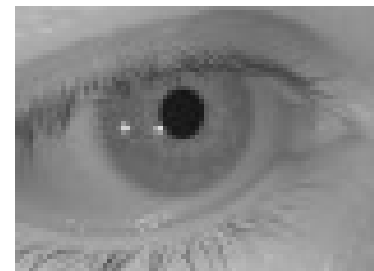
non-uniform illumination



poor resolution



off-angle



UAE Border Control System



- The **United Arab Emirates** (UAE) Ministry of Interior requires iris recognition tests on **foreigners entering UAE** from all air, land, and sea ports.
- Via internet links each traveler is compared against each of **1,000,000 expellees** (foreign nationals expelled for various violations), whose IrisCodes were registered in a central database upon expulsion.
- The time required for an exhaustive search through the database is about **1 second**.
- On an average day, 12,000 arriving passengers are compared against the entire watch list of 1,000,000 in the database; this is about **12 billion comparisons per day**.

Other Applications

- Aadhaar Program - **India**
- National ID Program – **Indonesia**
- National Population Register ("RENAPO") - **Mexico**
- CAC Program- **US** Department of Defense
- Privium Program - Schiphol Airport, **Netherlands**
- NEXUS and CANPASS program – **Canada**

Coal Mine in China



Iris on the Move

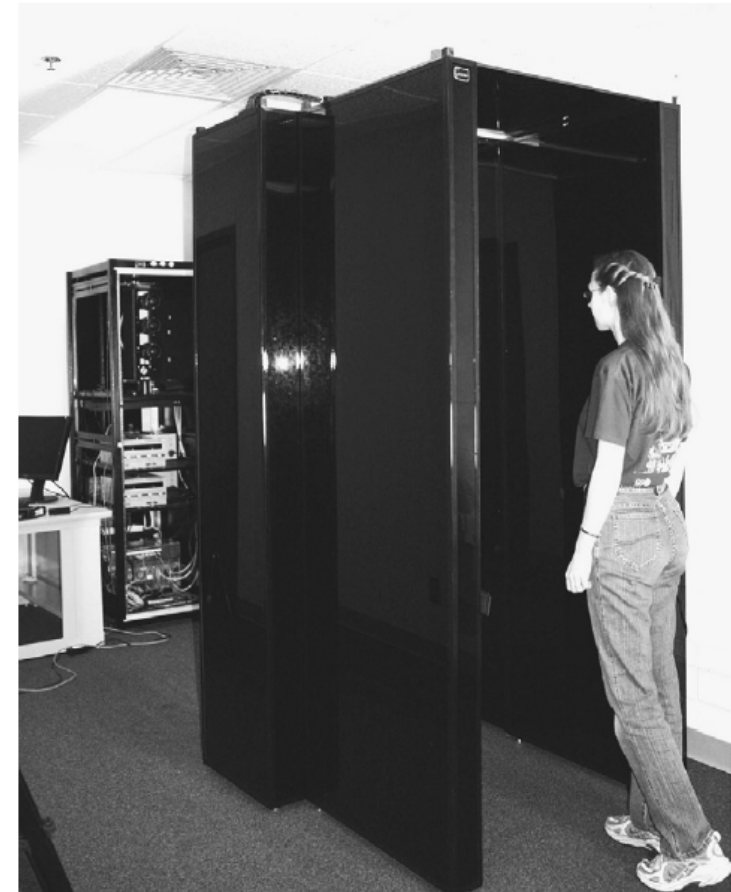
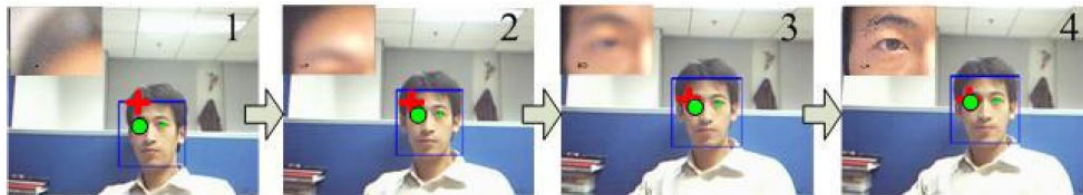
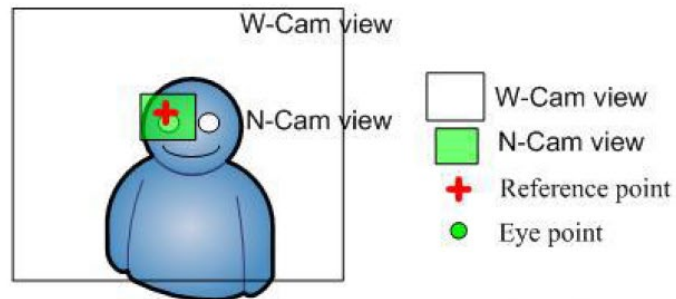
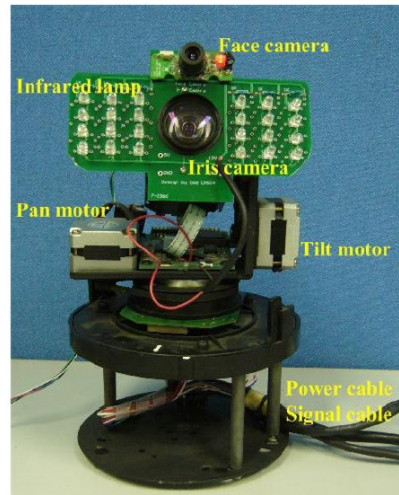
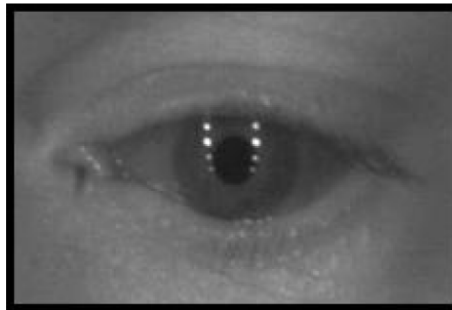
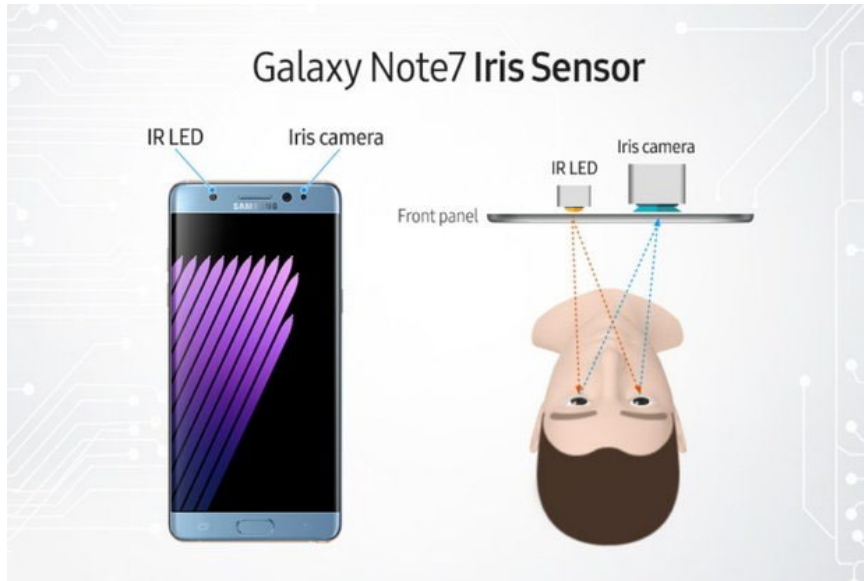


Fig. 4. This “Iris on the Move” portal acquires an iris image as subject walk through a portal at normal walking speed. The portal itself contain infrared lights to illuminate the subject. Three high-zoom video cameras in the far cabinet take video streams of the subject.

Iris on the Move



Smartphones: Near Infra-Red (NIR)





Iris: Open Directions

- Nowadays research on iris is active on:
 - Recognition at a distance
 - Surveillance scenarios
 - Mobile Applications
 - Fake iris detection, spoofing & template protection techniques

Outline

Overview of Main Biometric Modalities

- Fingerprint
- Face
- Iris

Other Biometric Modalities

Traits with Legacy Database



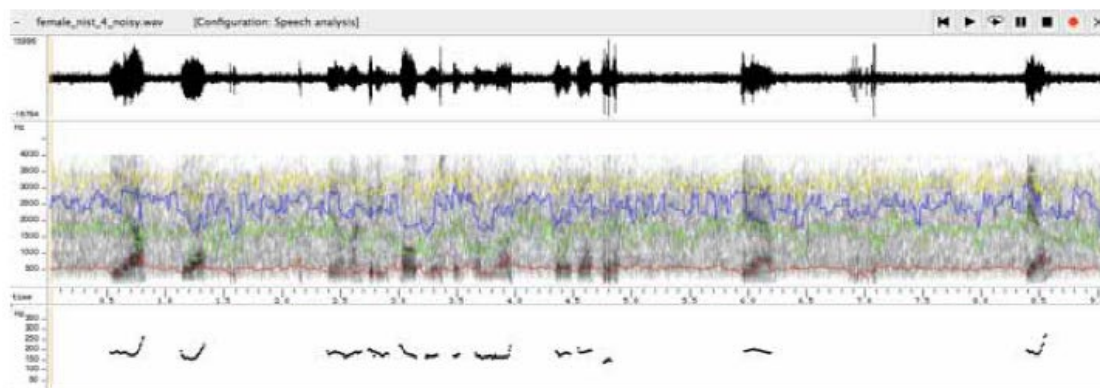
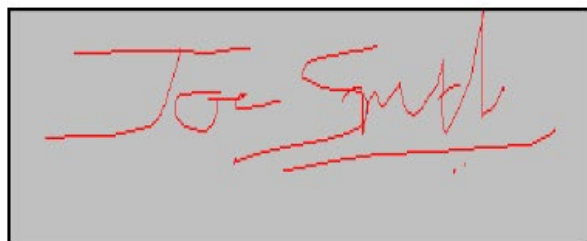
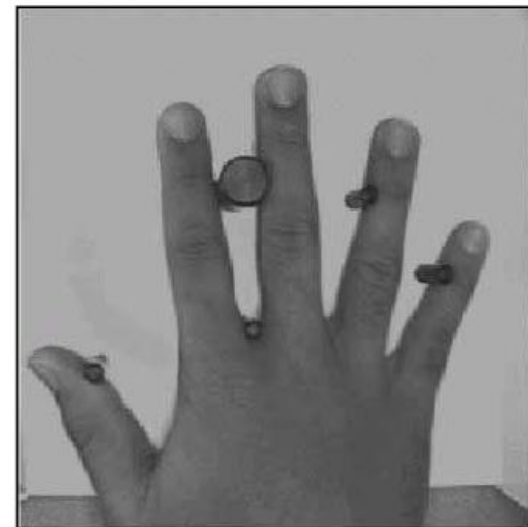
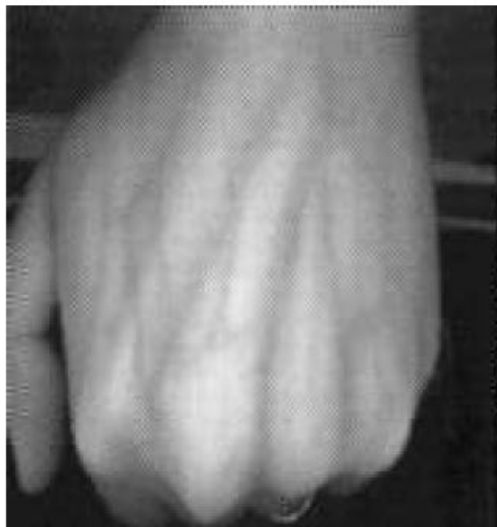
- ~30% of latents are of palmprints
- FBI's NGI will have latent palmprint matching capability



- Trace evidence
- Innocence project
- Real-time DNA matching

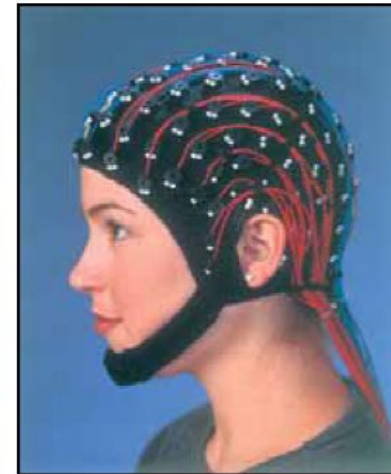
Capability to conduct 1:N search

Traits in Commercial Systems



Primarily for 1:1 match

Traits in Laboratory Stage



Traits in Laboratory Stage

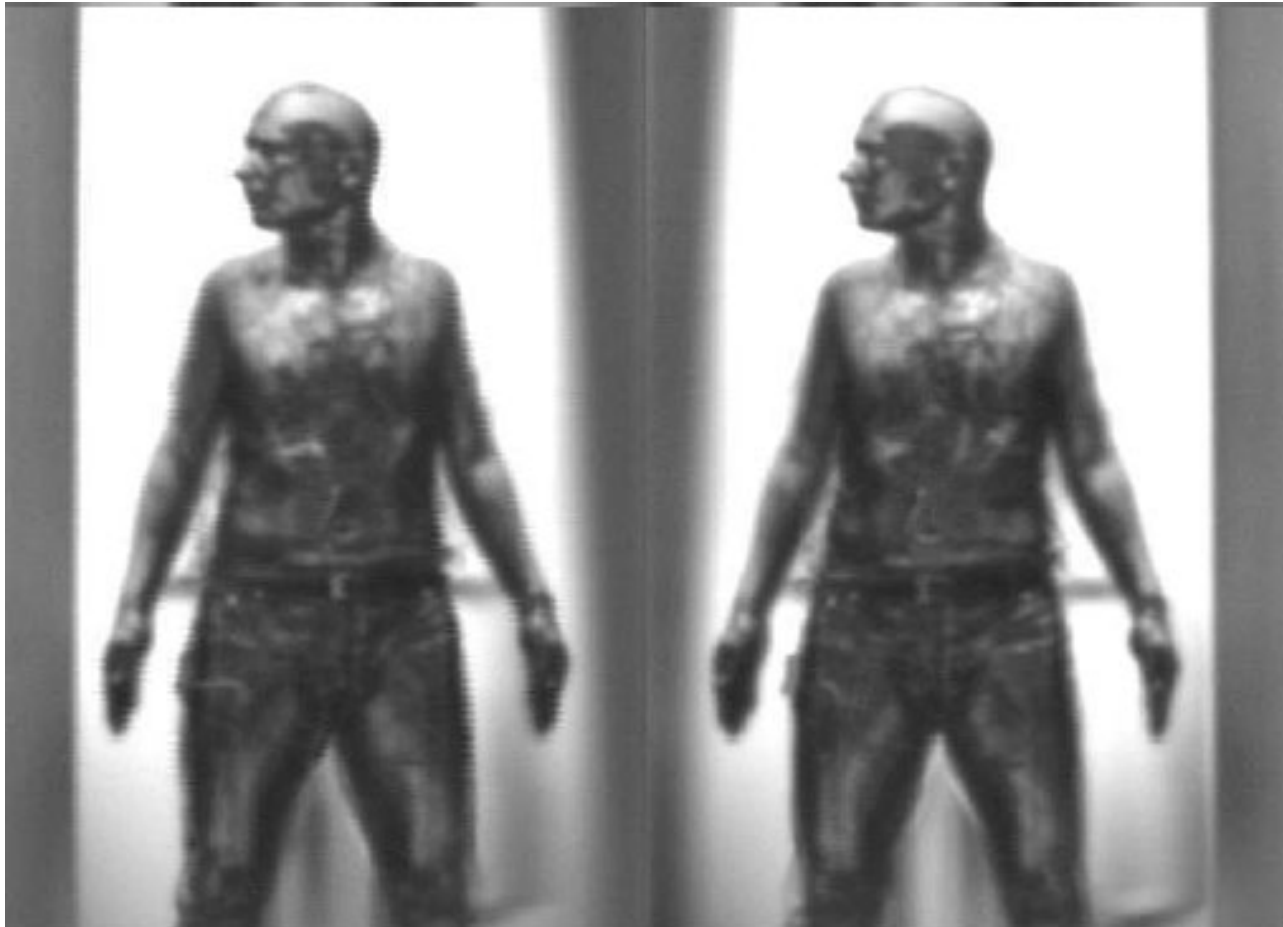
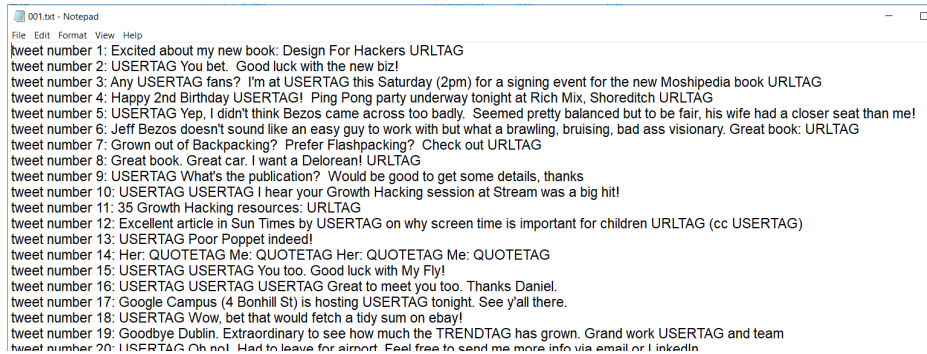


Fig. 4: Two screening images from the mmW TNO Database acquired with a stereo radiometric scanner. A screening image

Traits in Laboratory Stage

- Identification of author of **Tweets!**

- Features: frequency of letters, punctuation, numbers, hashtags, usetags...
- Challenge: the available text is too short (so several tweets are combined)



001.txt - Notepad

File Edit Format View Help

tweet number 1: Excited about my new book: Design For Hackers URLTAG

tweet number 2: USERTAG You bet. Good luck with the new biz!

tweet number 3: Any USERTAG fans? I'm at USERTAG this Saturday (2pm) for a signing event for the new Moshpedia book URLTAG

tweet number 4: Happy 2nd Birthday USERTAG! Ping Pong party underway tonight at Rich Mix, Shoreditch URLTAG

tweet number 5: USERTAG Yep, I didn't think Bezos came across too badly. Seemed pretty balanced but to be fair, his wife had a closer seat than me!

tweet number 6: Jeff Bezos doesn't sound like an easy guy to work with but what a brawling, bruising, bad ass visionary. Great book: URLTAG

tweet number 7: Grown out of Backpacking? Prefer Flashpacking? Check out URLTAG

tweet number 8: Great book. Great car. I want a Delorean! URLTAG

tweet number 9: USERTAG What's the publication? Would be good to get some details, thanks

tweet number 10: USERTAG USERTAG I hear your Growth Hacking session at Stream was a big hit!

tweet number 11: 35 Growth Hacking resources: URLTAG

tweet number 12: Excellent article in Sun Times by USERTAG on why screen time is important for children URLTAG (cc USERTAG)

tweet number 13: USERTAG Poor Poppet indeed!

tweet number 14: Her: QUOTETAG Me: QUOTETAG Her: QUOTETAG Me: QUOTETAG

tweet number 15: USERTAG USERTAG You too. Good luck with My Fly!

tweet number 16: USERTAG USERTAG USERTAG Great to meet you too. Thanks Daniel.

tweet number 17: Google Campus (4 Bonhill St) is hosting USERTAG tonight. See y'all there.

tweet number 18: USERTAG Wow, bet that would fetch a tidy sum on ebay!

tweet number 19: Goodbye Dublin. Extraordinary to see how much the TRENDTAG has grown. Grand work USERTAG and team

tweet number 20: USERTAG Oh no! Had to leave for airport. Feel free to send me more info via email or LinkedIn

tweet number 1: Excited about my new book: Design For Hackers URLTAG

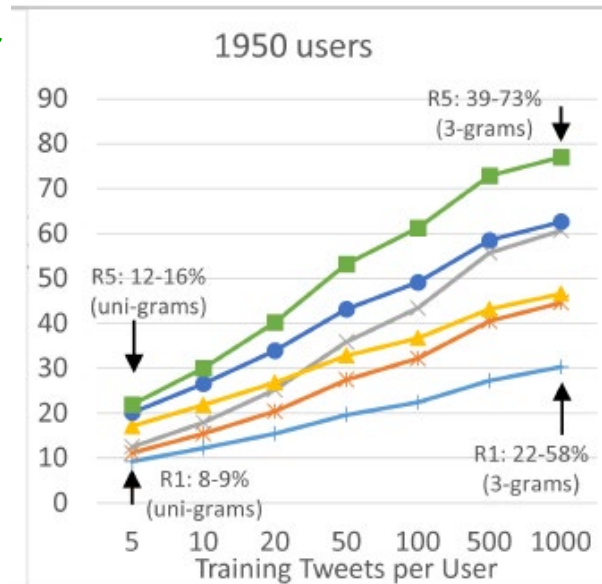


Vector of 150 elements per tweet

1 0 0 0 38 37 4 33 14 1 13 0 8 1 2 2 2 2 5 1 1 1 2 0 2 0 1 2 4 0 0 2 2 2 1 0
1 1 1 0 0 0 0 1 1 1 0 1 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 2 2 2 1 4 0 1 0 2 0
2 0 1 2 4 0 0 2 2 2 1 0 1 1 1 0 2 5 2 4 1 0 1 0 0 0 2 4 2 4 1 0 0 0 0 0 0 0 0 0
0 1 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0

- Rank-1 with **5 tweets** per user: 8-9%

- Rank-1 with **1000 tweets** per user: 22-58%



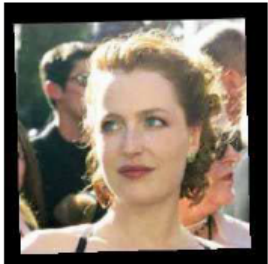
F. Alonso-Fernandez, N. Belvisi, K. Hernandez-Diaz, N. Muhammad, J. Bigun. "Forensic Writer Identification Using Microblogging Texts", *IEEE Transactions on Biometrics, Behavior, and Identity Science*, accepted 2021

Soft-Biometrics

- Semantical descriptions as made by humans
- May not possess sufficient distinctiveness, but can be useful to enhance recognition under certain conditions, e.g. surveillance
- Challenges
 - How to estimate them automatically from e.g. face, iris, gait...
 - How to use such information in the context of recognition

Success cases of genuine comparisons: genuine comparisons detected as impostor by the face verification system but as genuine by the fusion system

A) Gillian Anderson



Caucasian
Youth
Female
No Glasses
No Beard
No Moustache

score_face -0.18 score_soft 0.8069

B) Patty Schnyder

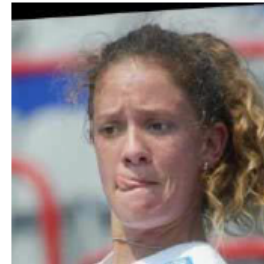


Caucasian
Youth
Female
No Glasses
No Beard
No Moustache



Caucasian
Youth
Female
No Glasses
No Beard
No Moustache

score_face 0.0024 score_soft 0.8069



Caucasian
Youth
Female
No Glasses
No Beard
No Moustache

C) Sandra Bullock



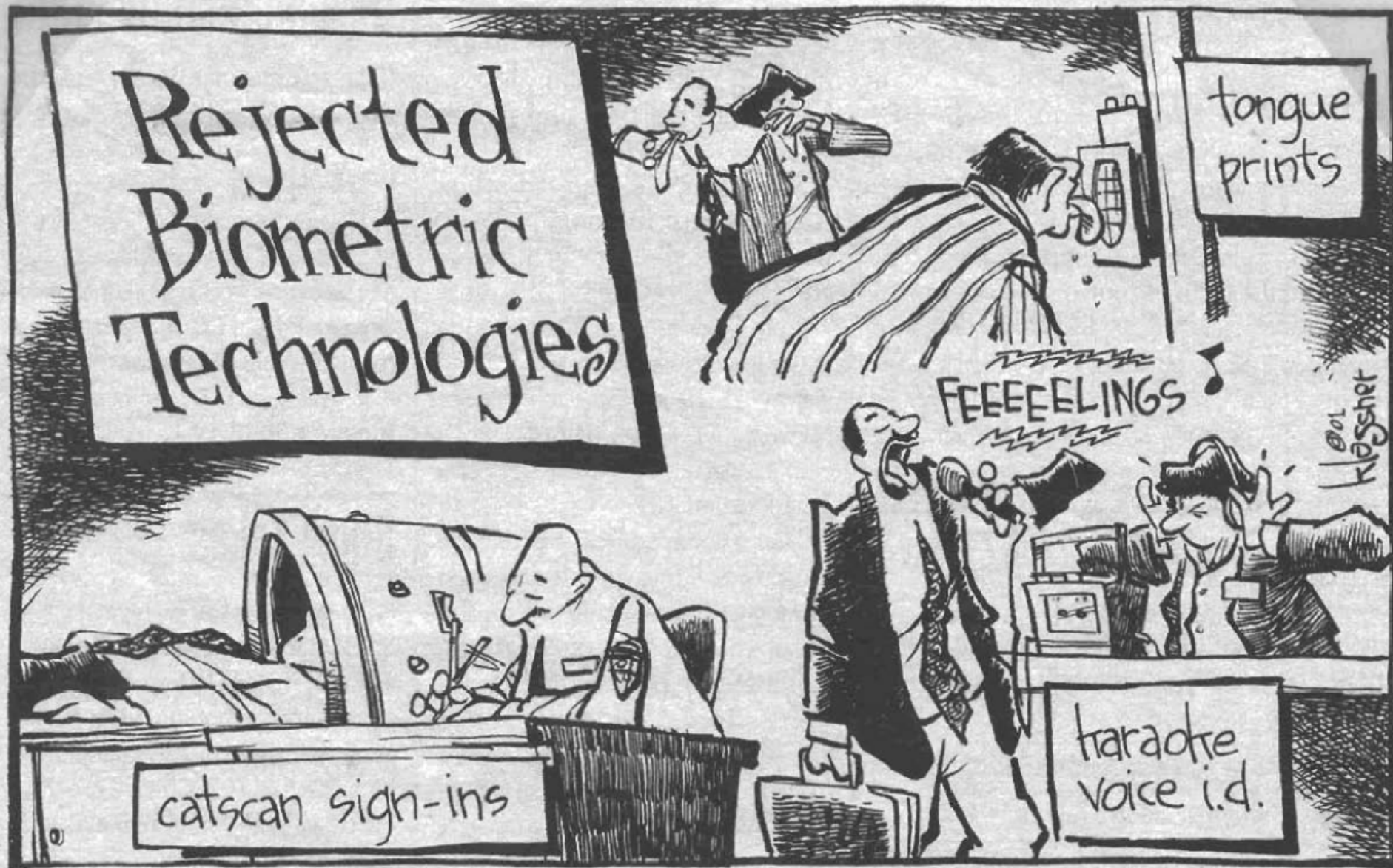
Caucasian
Middle Aged
Female
No Glasses
No Beard
No Moustach

score_face -0.3023 score_soft 0.8069



Caucasian
Middle Aged
Female
No Glasses
No Beard
No Moustach

Rejected Biometric Traits



Introduction to Biometric Technologies and Applications (IV)

Fernando Alonso-Fernandez (F510)
ITE School

Email: feralo@hh.se

Course: Biometric Recognition
DI4025